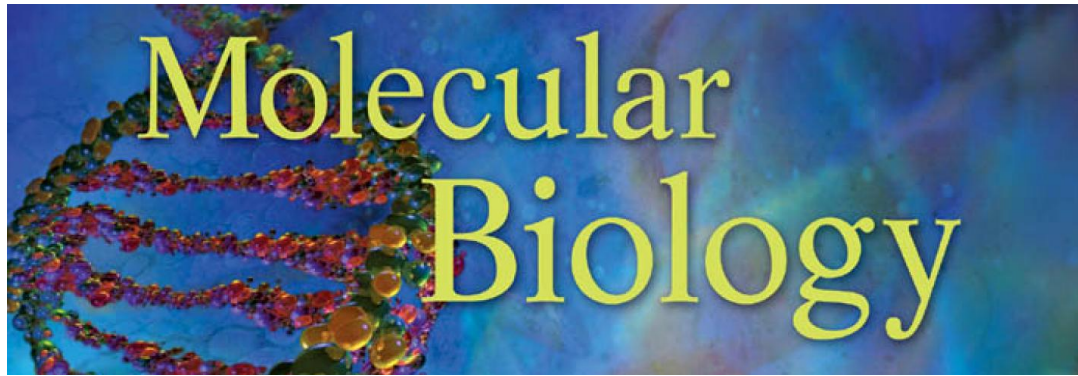




Gene structure and transcription in prokaryotes

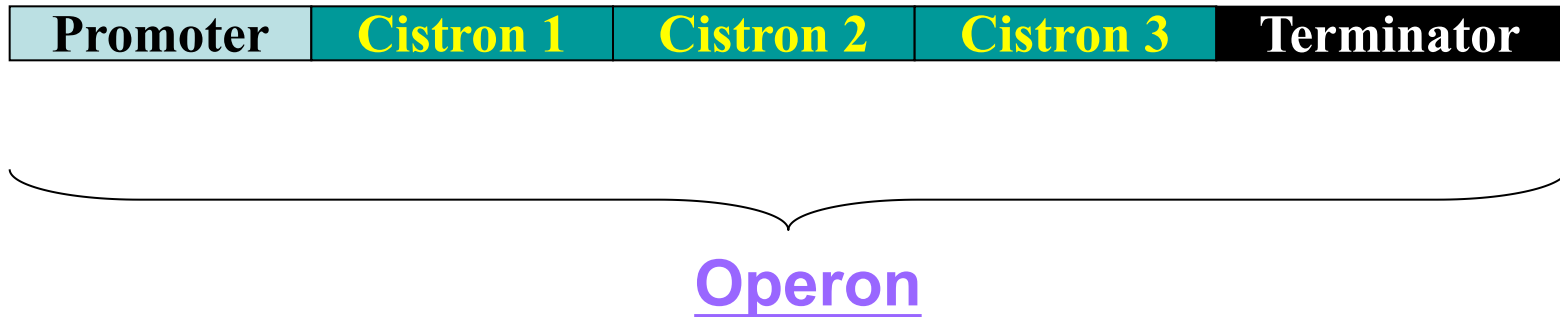
周明

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mingzhou@zju.edu.cn

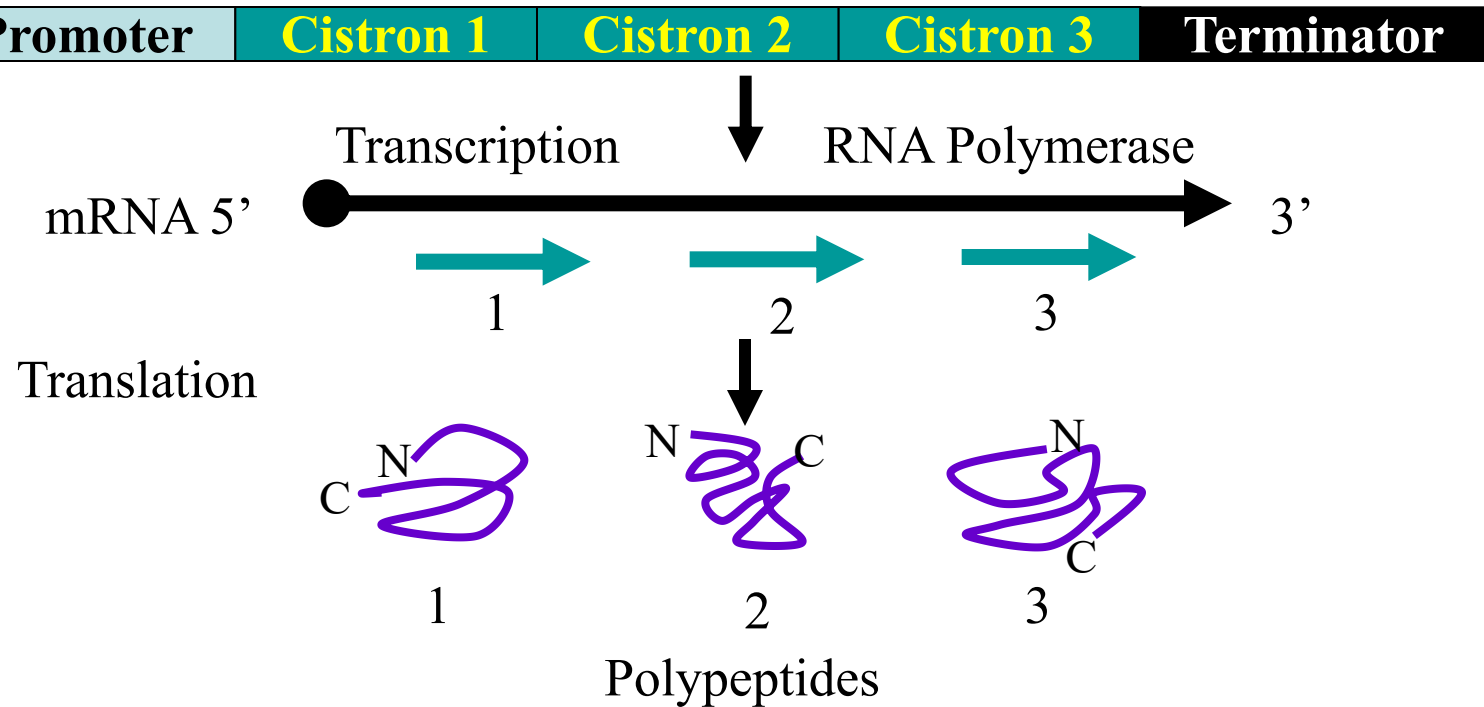
- 1. Prokaryotic Gene Structure**
- 2. The mechanism of transcription in Bacteria**
- 3. Operons: Fine control of bacterial transcription**

1. Prokaryotic Gene Structure



- Operon (操纵元) : a unit of gene expression in prokaryotes and which includes structural genes and control elements
- Structural **gene clusters** are **coordinately** controlled

1.1 Prokaryotic Gene Structures



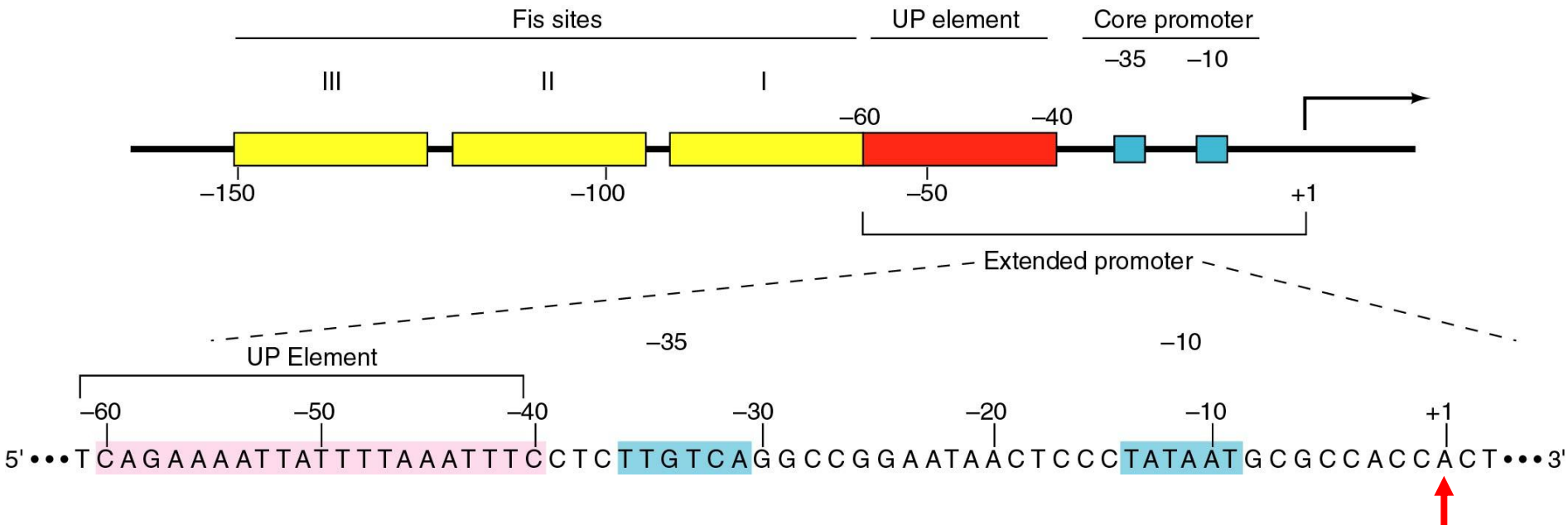
Cistron : (顺反子) **A section of DNA that contains the genetic code for a single polypeptide and functions as a hereditary unit.**

• **Polycistronic message** (多顺反子): **an mRNA bearing information from more than one gene**

1.2 Promoter structure in prokaryote

- Promoter (启动子): A DNA sequence to which RNA polymerase binds prior to initiation of transcription-usually located just upstream of the transcription start site of a gene.
- Two core promoter elements:
 - 10 Box (Pribnow box) : **TATAAT**, centered in -10 position.
 - 35 box: **TTGACA**, centered in -35 bp position.
- UP element: some promoter has an extra element in -40~-60.
- The closer to the consensus sequence, the stronger that promoter will be.

The *rrnB* P1 promoter



Fis sites: bacterial enhancers, which binds to transcription-activator protein Fis

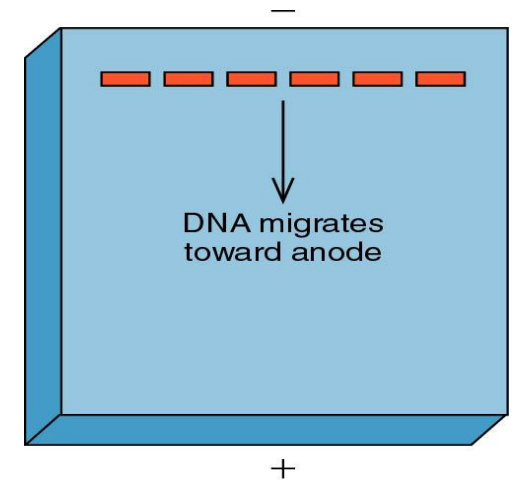
1.3 Transcription Initiation site

- The transcription initiation site is a purine in 90% of genes
- More A>G
- Mostly there are C and T bases on either side of the start site: i.e. C**G**T or C**A**T

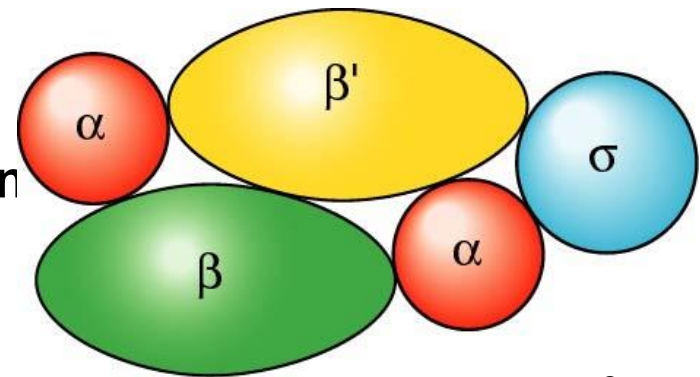
2. The mechanism of transcription in bacteria

2.1 RNA polymerase

- Isolated in 1969 by SDS- PAGE (polyacrylamide gel electrophoresis)
- Contains:
 1. Two large subunits (β, β')
 2. σ & α subunits
 3. RNA polymerase holoenzyme (全酶) includes β', β, σ & α_2
 4. Core enzyme (核心酶) : without σ subunit

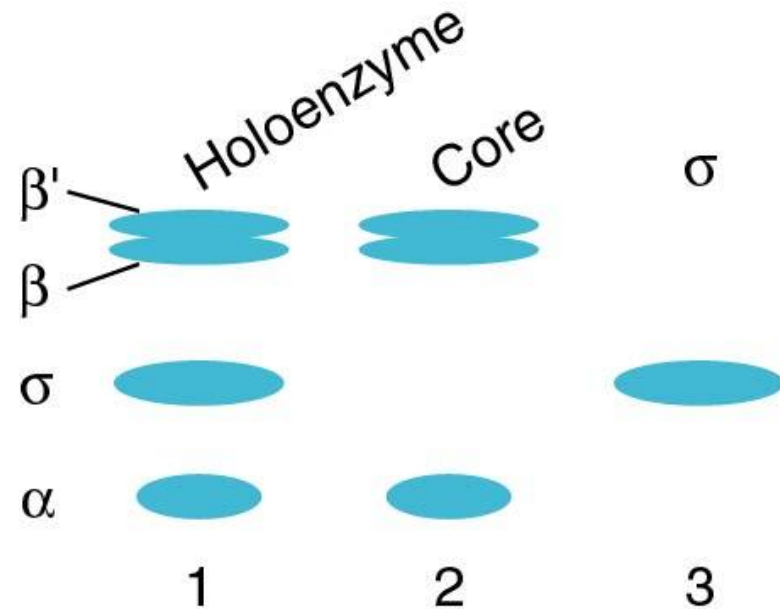


Gel Electrophoresis



Core enzyme and σ factor

- Core enzyme contains the basic transcription machinery
- Core enzyme only transcribe nicked DNA template, not intact DNA
- Without σ (sigma) factor, the core enzyme lost specificity of transcription Initiation site



2.2 Transcription Initiation

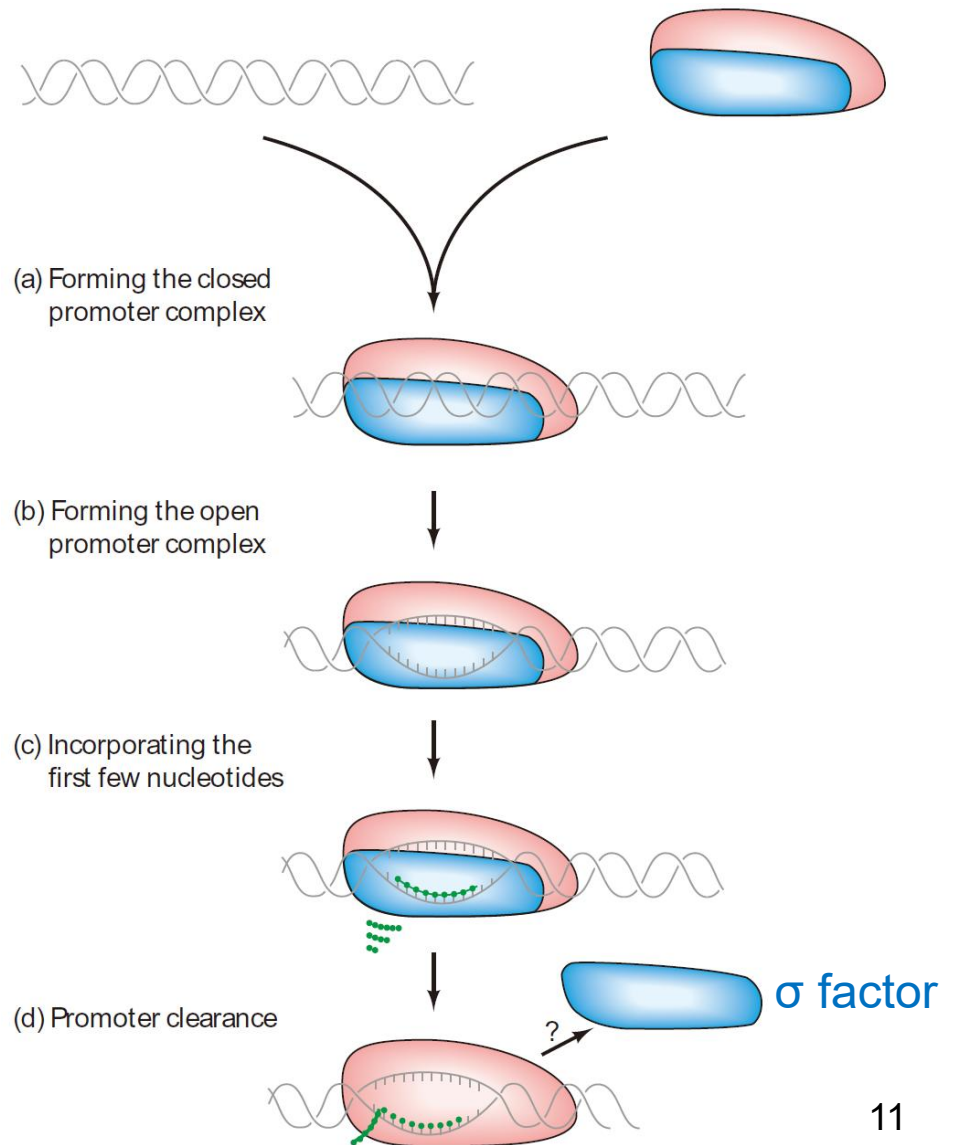
Holoenzyme search for a binding site on the DNA

(a) Forming the closed promoter complex

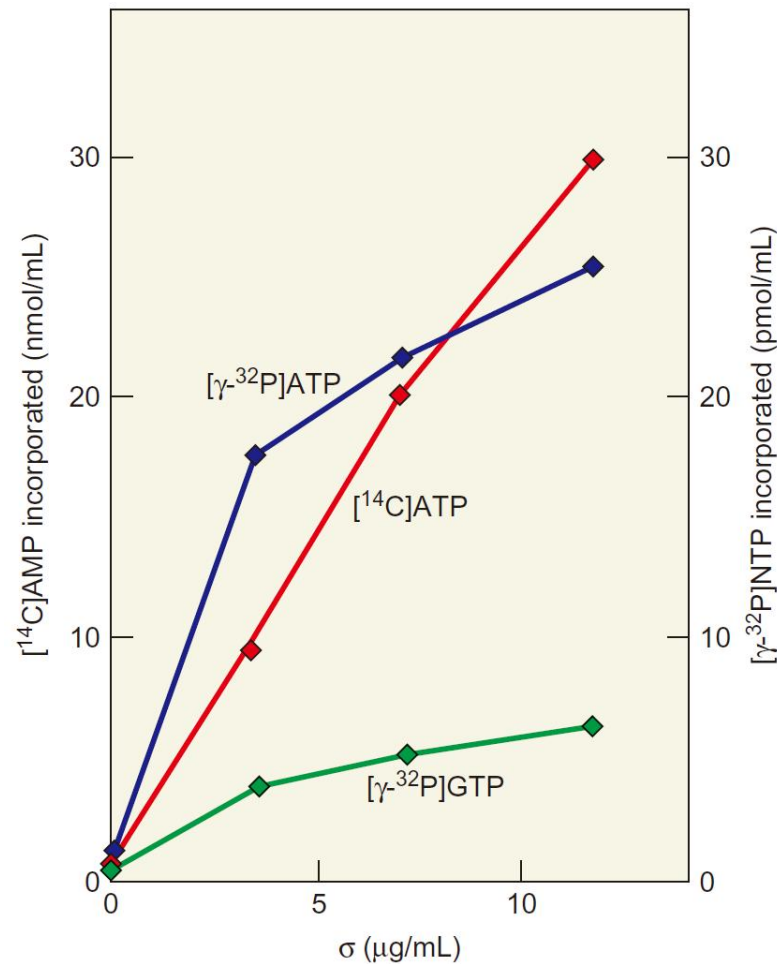
(b) Bounding tightly, melting DNA and forming an open promoter complex

(c) Polymerase remains at the promoter, in an initial transcribing complex

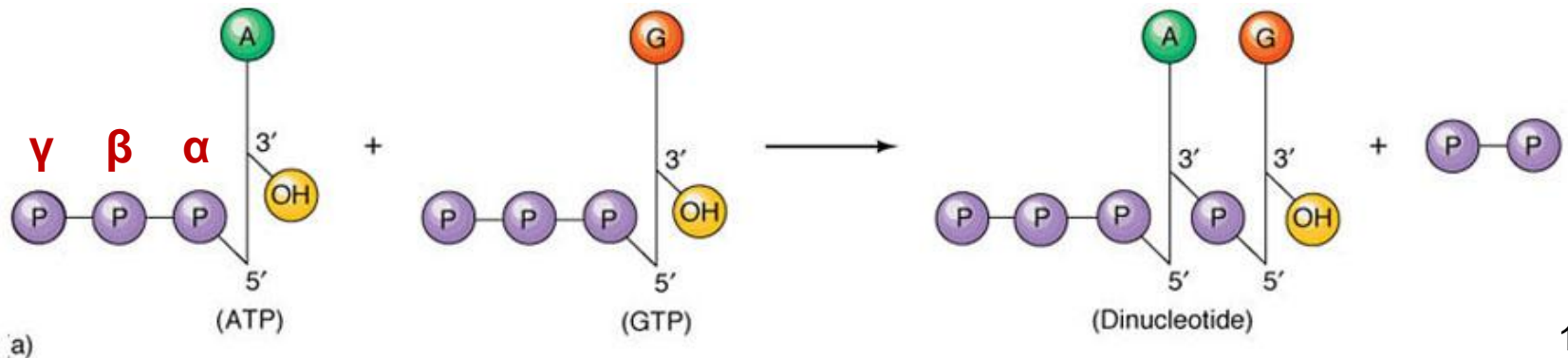
(d) Promoter clearance

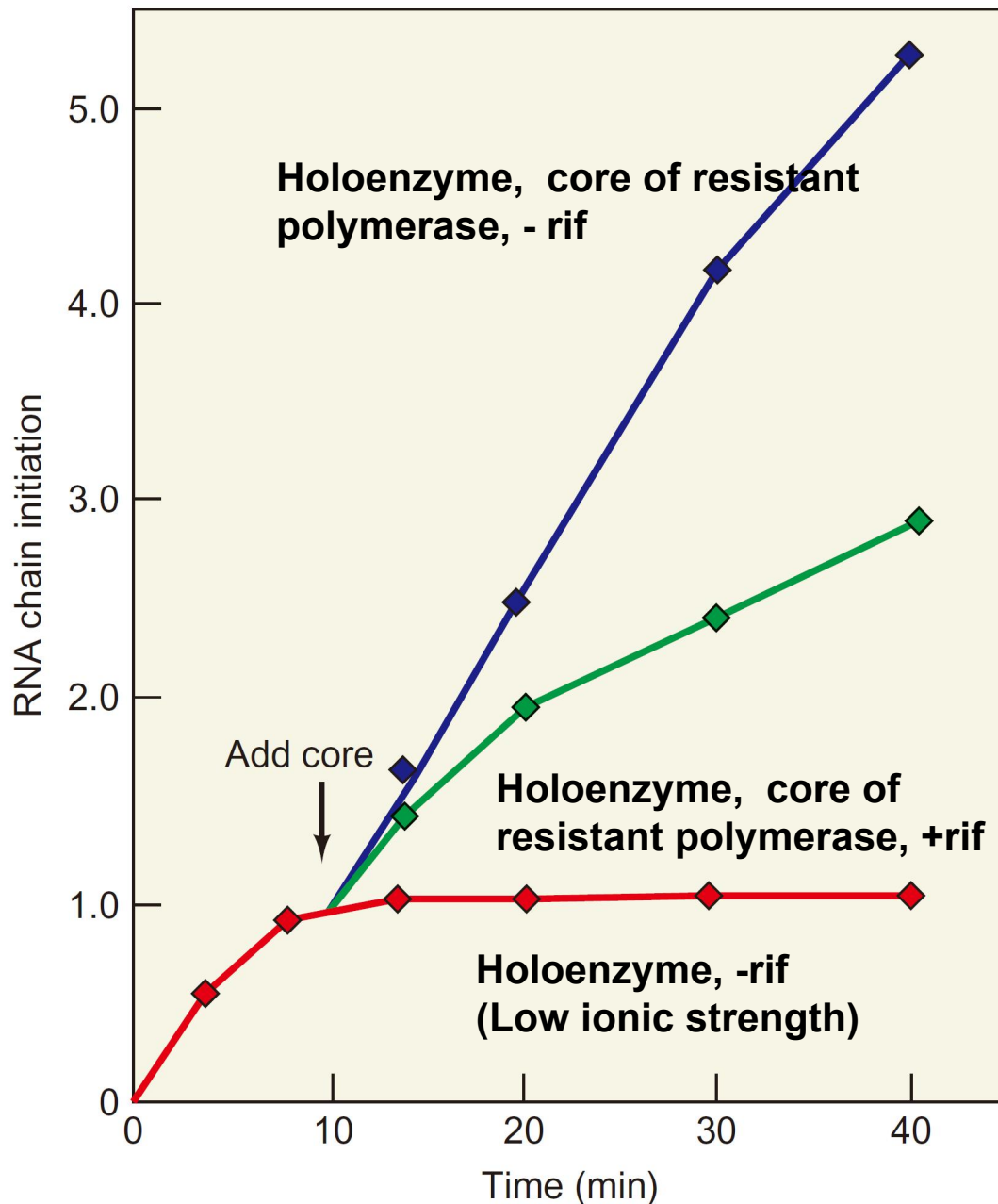


sigma stimulate initiation, but not elongation, of transcription



1. transcription initiates in A or G, more A than G
2. Measure the rate of initiation by using $\gamma^{32}\text{P}$ labeled ATP or GTP





σ factor can be reused

Measure the rate of initiation by using $\gamma^{32}\text{P}$ labeled ATP and GTP

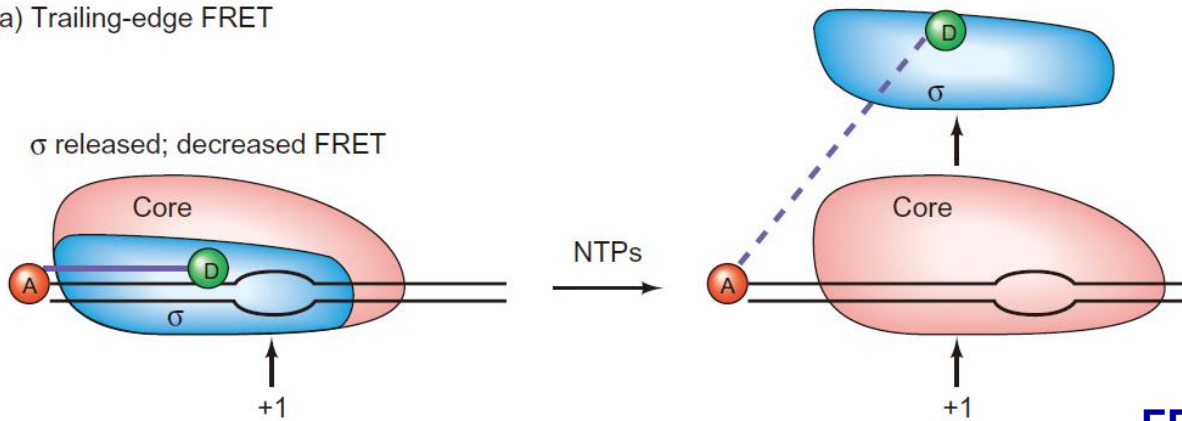
Result: sigma can be reused, and the core, not σ factor, governs rifampicin sensitivity or resistance.

σ factor may not dissociate from core during elongation

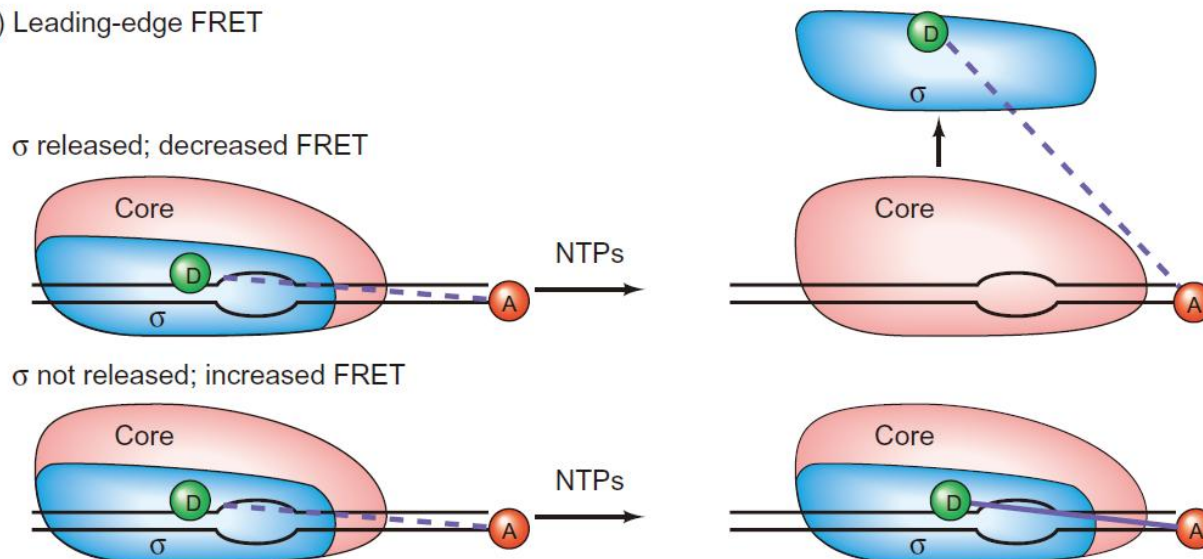
FRET: fluorescence resonance energy transfer

- A fluorescence donor (D, green) is attached to the single cysteine residue in a $\sigma 70$ mutant
- A fluorescence acceptor (A, red) is attached to the 5'-end of the DNA.
- FRET efficiency depends on the distance between the donor and receptor

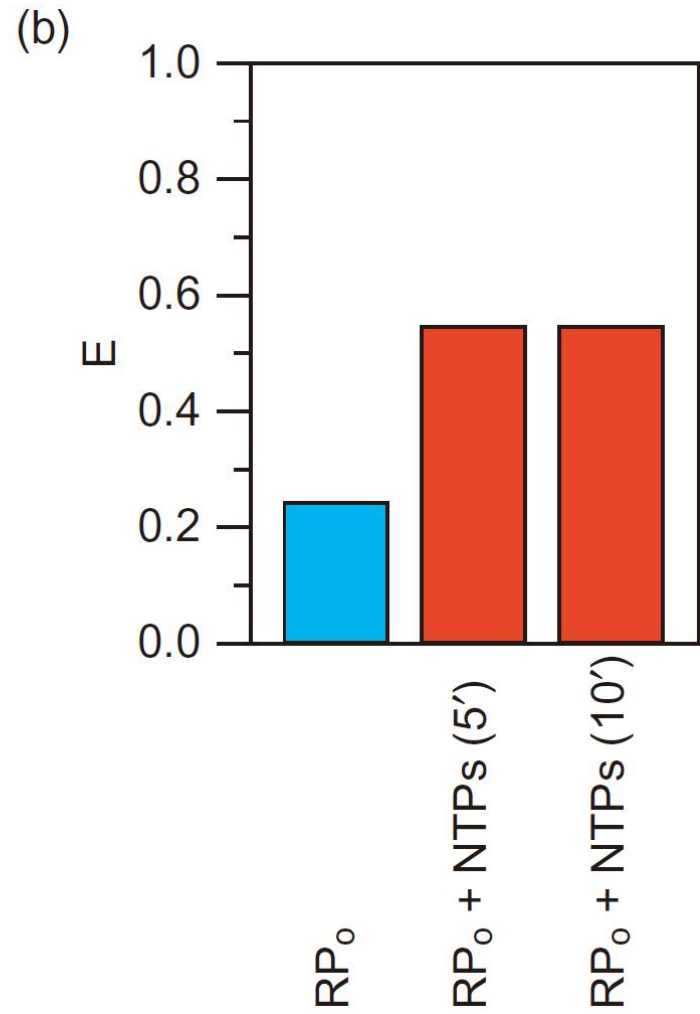
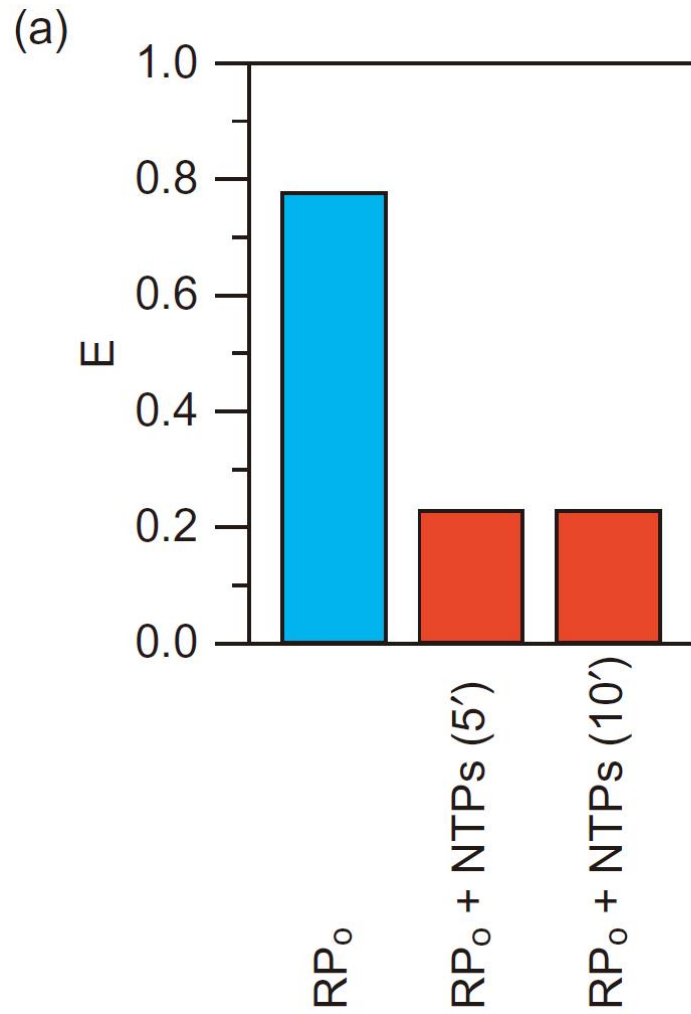
(a) Trailing-edge FRET



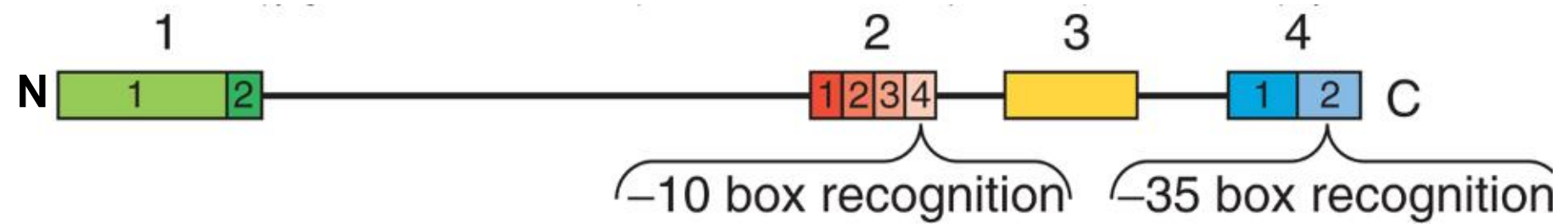
(b) Leading-edge FRET



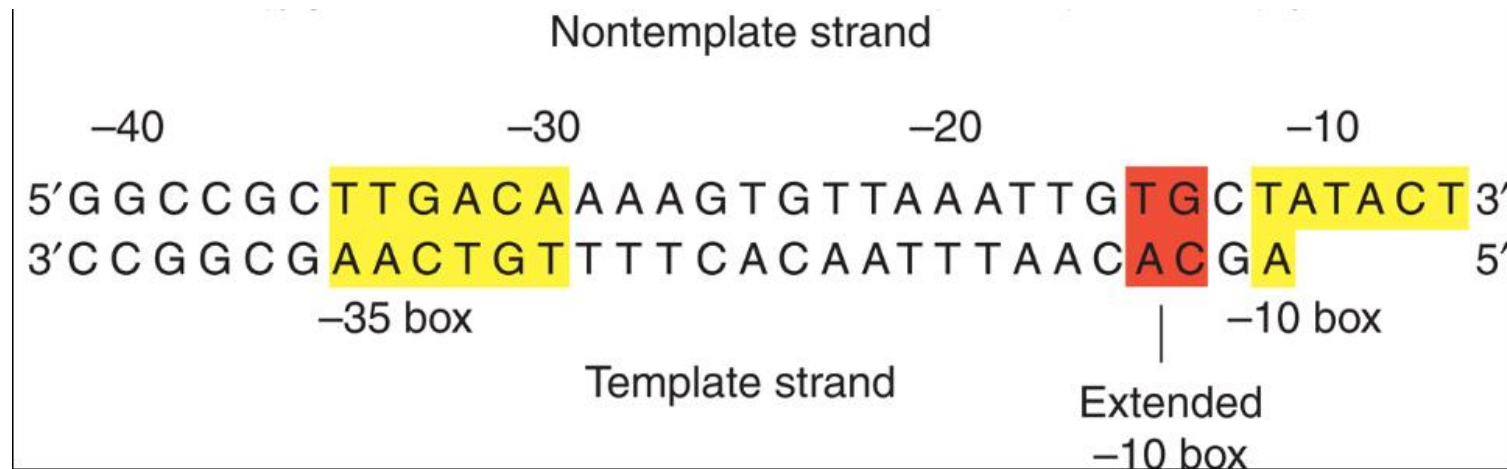
Results of FRET



Two conserved regions of σ are involved in promoter -10 box and -35 box recognition

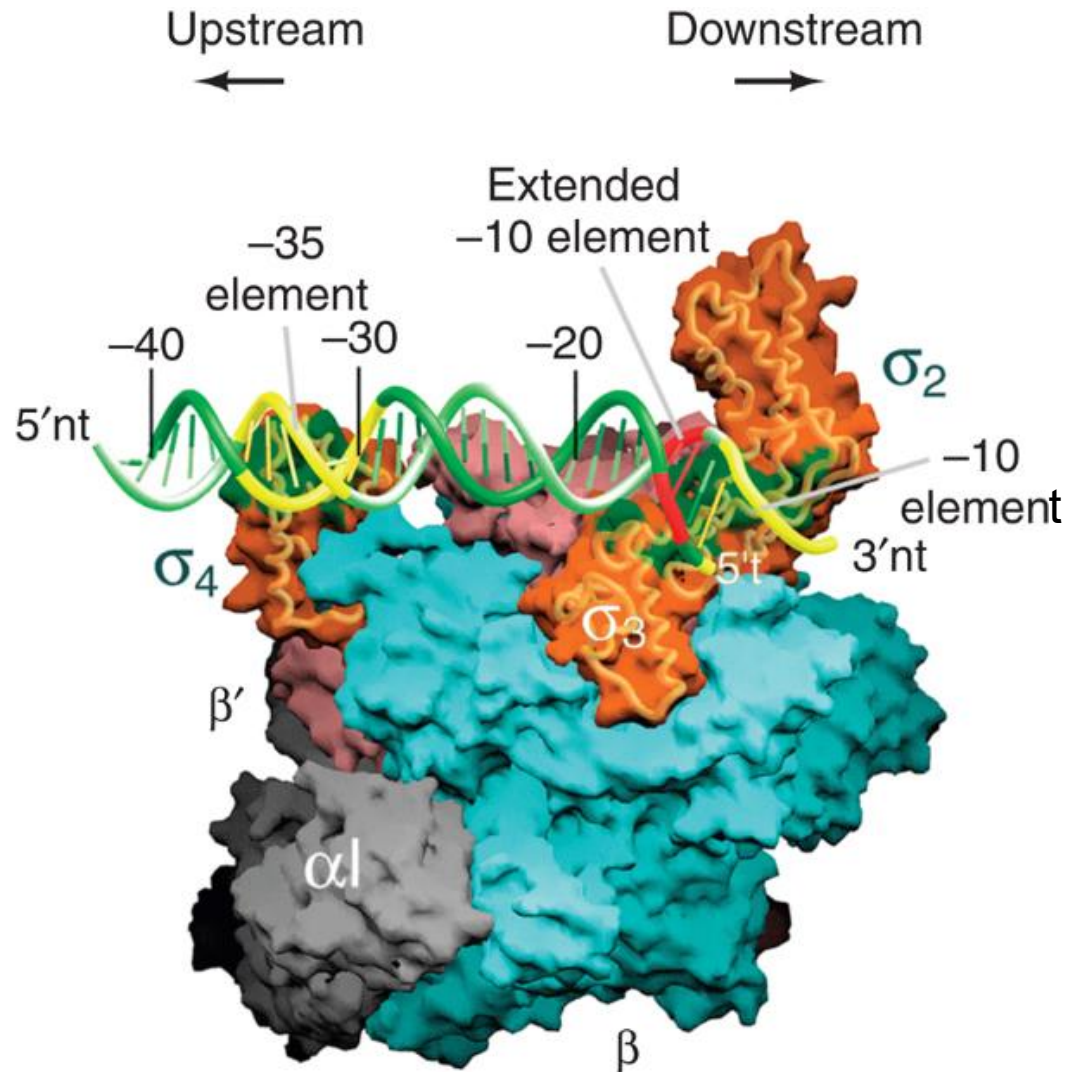


Up: four regions of σ factor

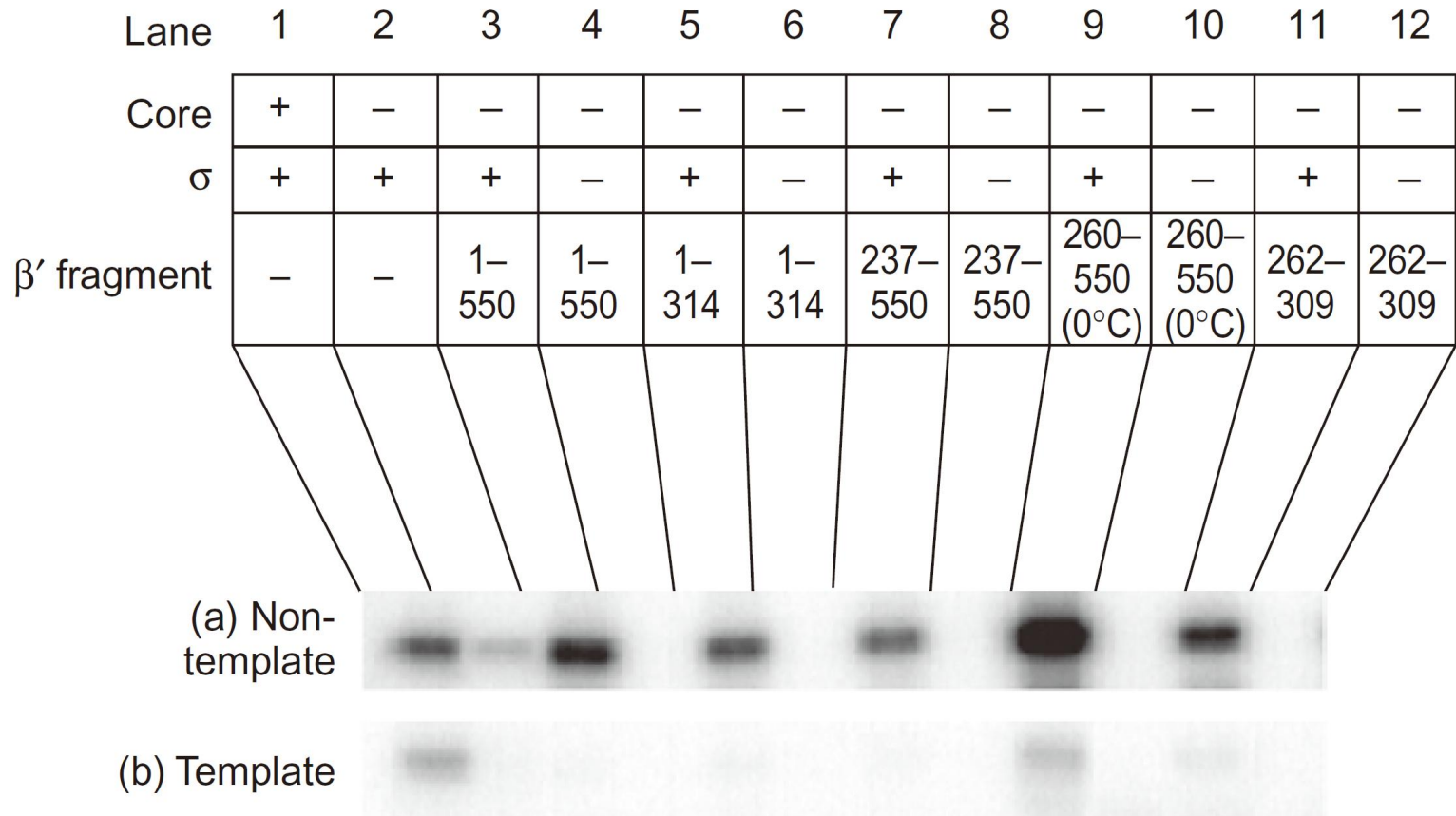


down: promoter region of template DNA

Structure of the open promoter complex



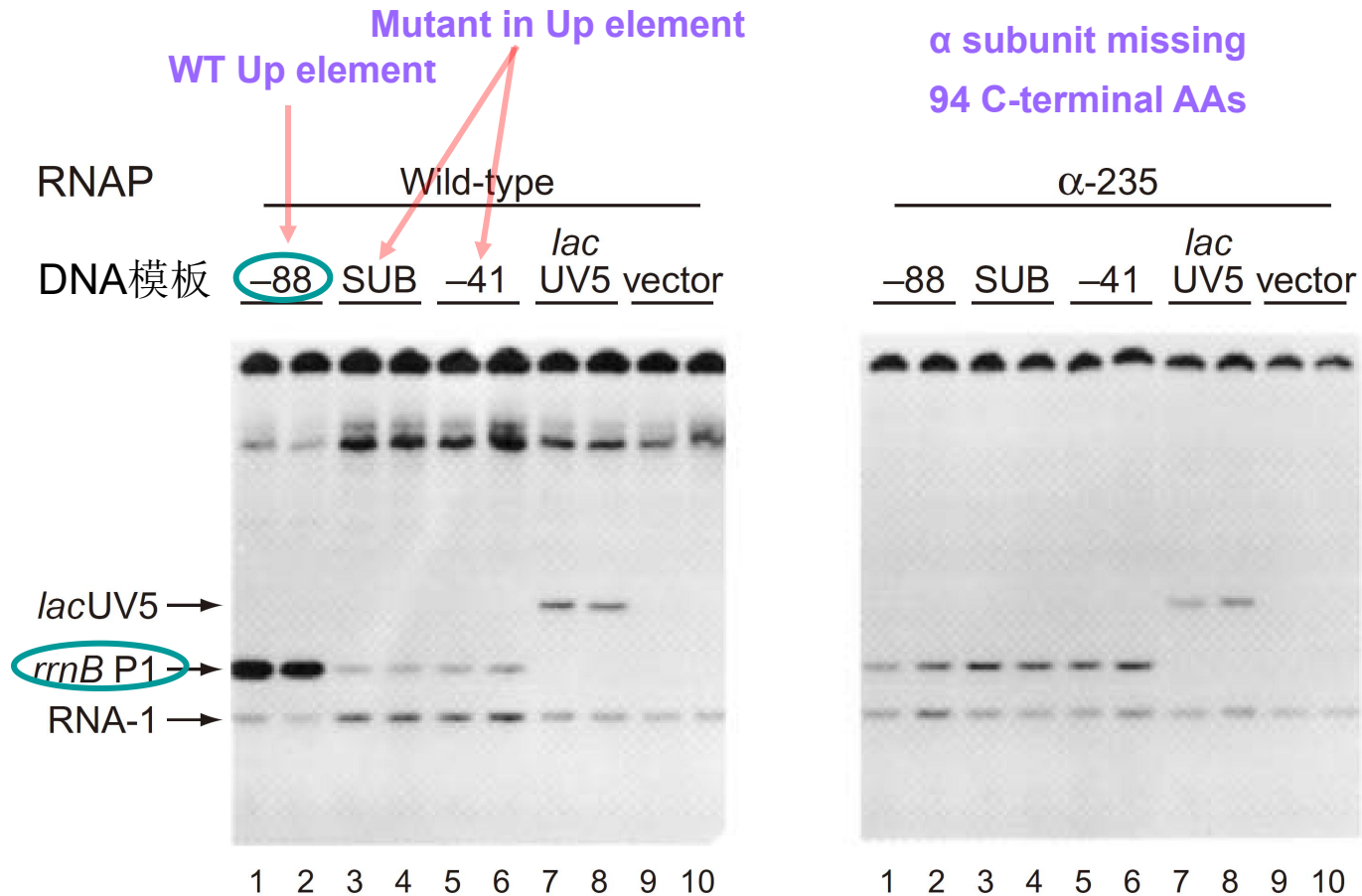
β' subunit Stimulates σ binding to the nontemplate strand in the -10 region of the promoter



Summary for σ factor

- (1) σ factor is involved in promoter -10 box and -35 box recognition, so σ factor can select the transcriptional initiation site.
- (2) The σ factor by itself cannot bind to DNA, but interaction with core unmasks a DNA-binding region of σ .
- (3) After the initiation, σ factor is bound loosely in the elongating state.

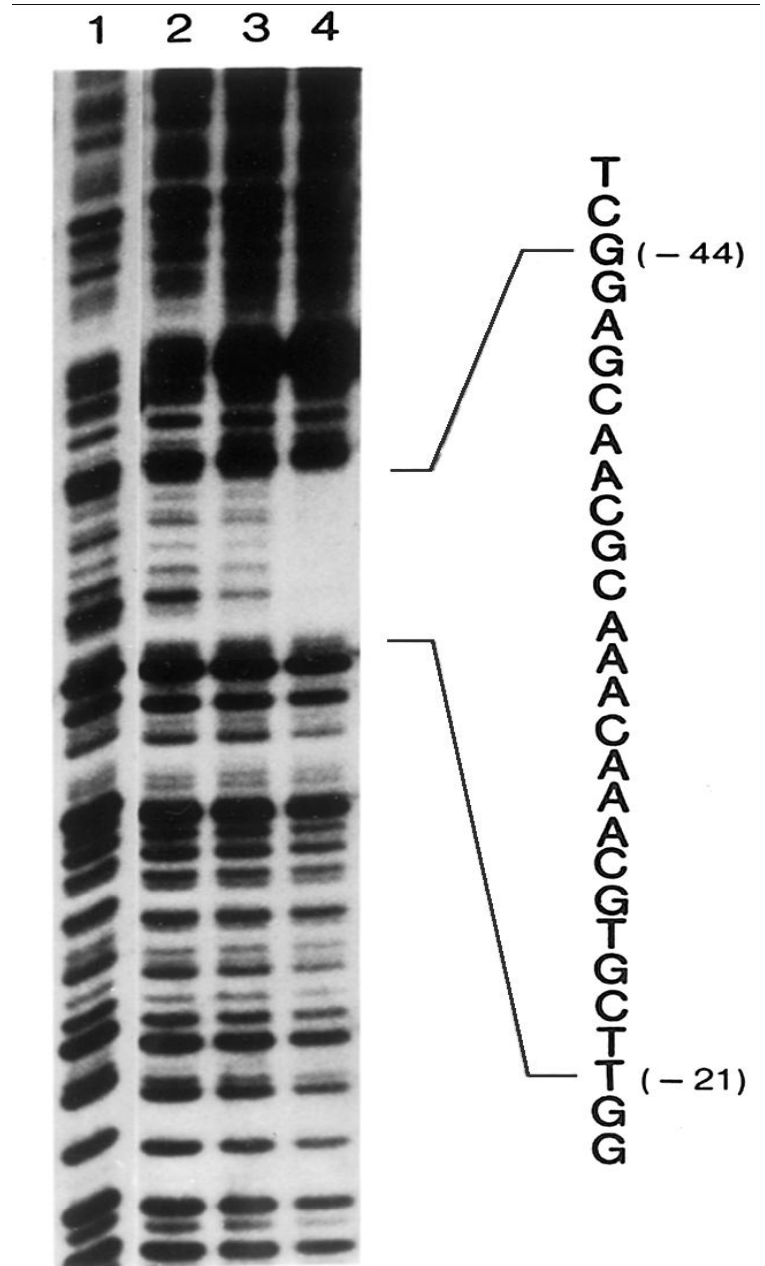
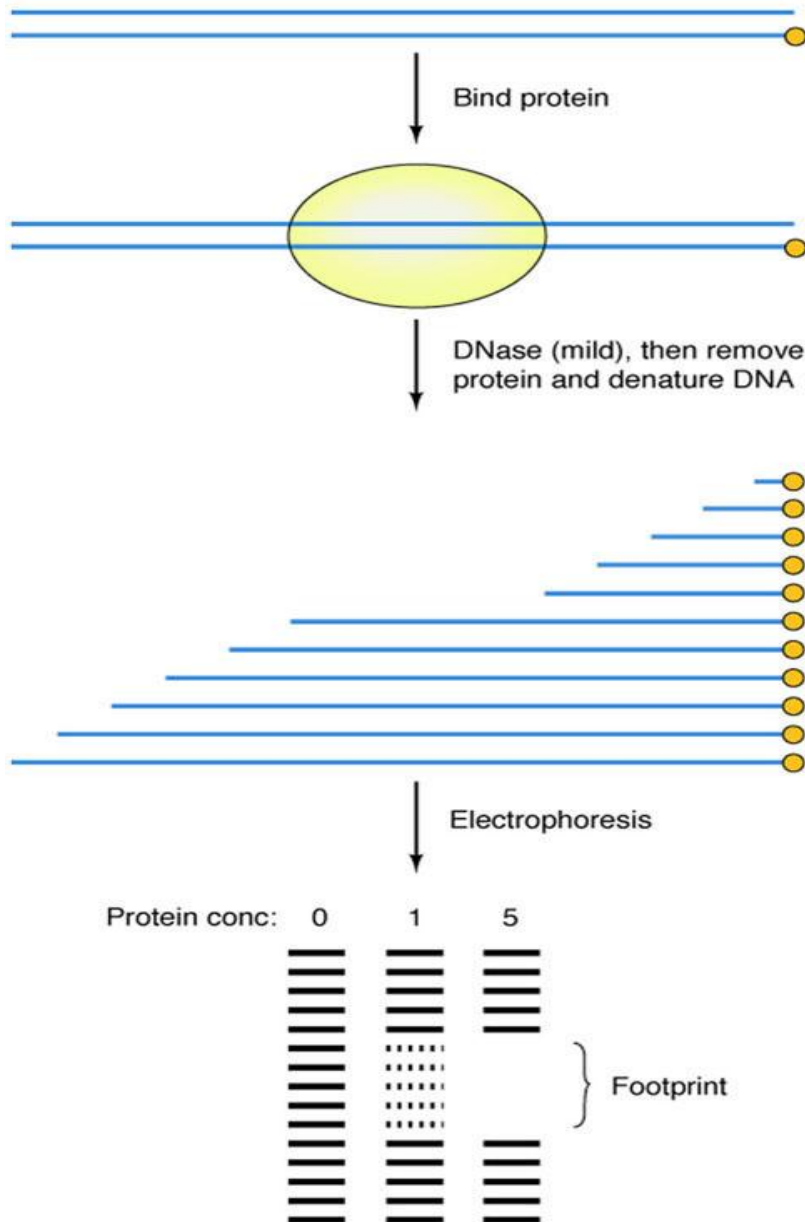
The role of the α subunit in UP element recognition

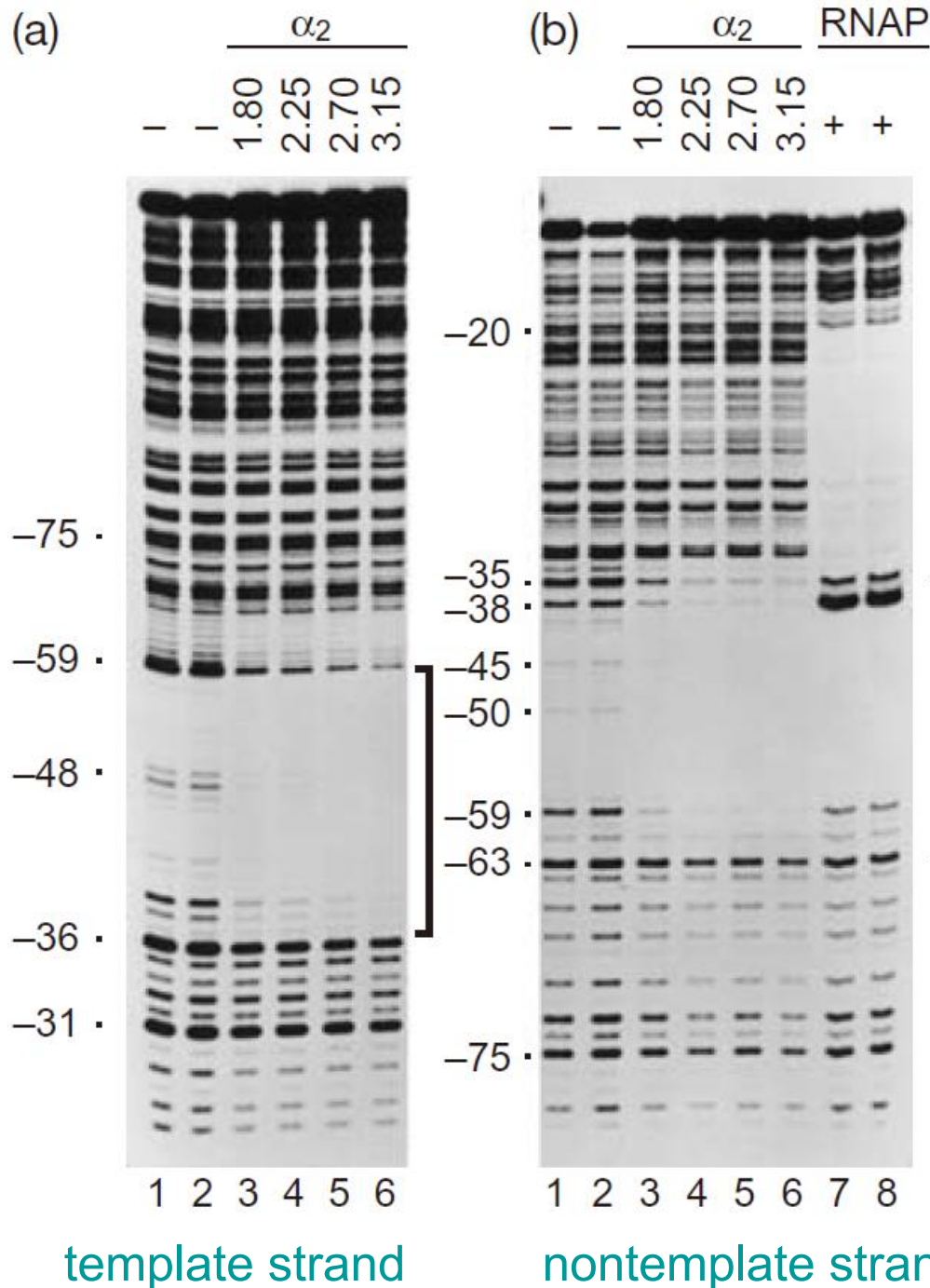


W/O a functional α subunit, UP element is no longer an enhancer

DNase footprinting

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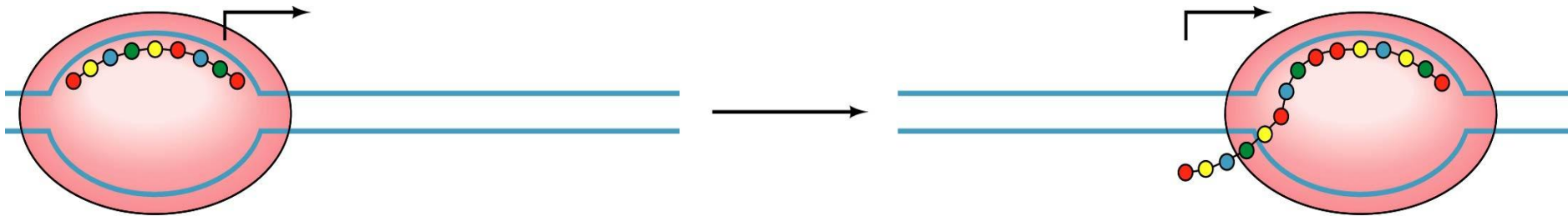




binding between α subunit & UP element

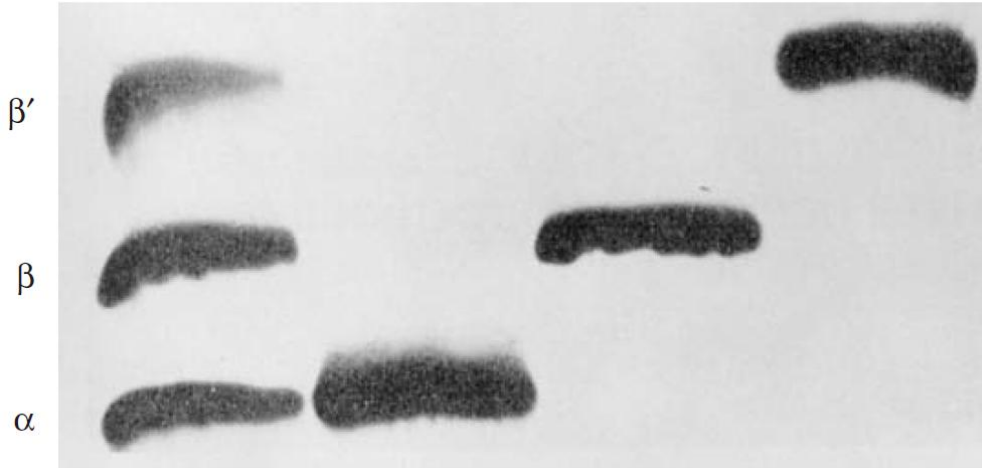
2.3 Elongation of transcription

- Need core enzyme
- DNA unwinding: topoisomerases

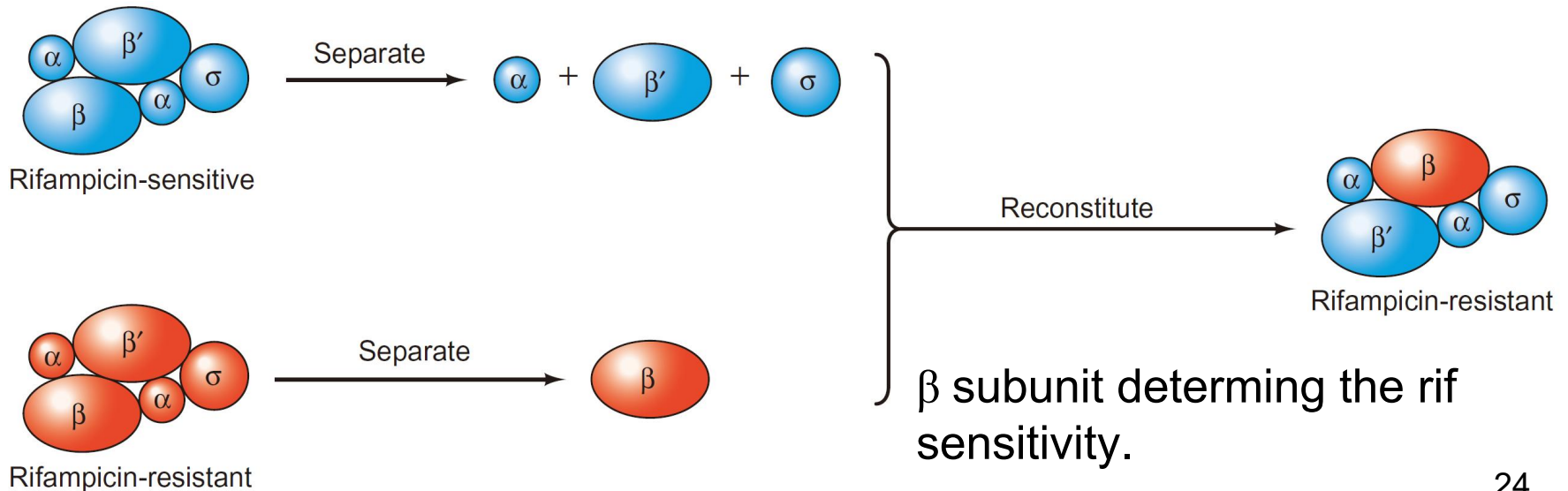


β in Phosphodiester bond formation

1 2 3 4



@ Urea was used to denature the core enzyme, the denatured protein can be re-natured by removing urea.



β in Phosphodiester bond formation

1. Rif blocks the transcription initiation

β subunit determining the rif sensitivity

2. Streptodigin (利链菌素) blocks RNA chain elongation

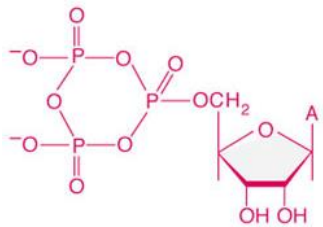
β subunit determining the Streptodigin sensitivity

3. The common thing for initiation and elongation is phosphodiester bond formation.

β subunit ensures the formation of phosphodiester bond

β in Phosphodiester bond formation

(a)

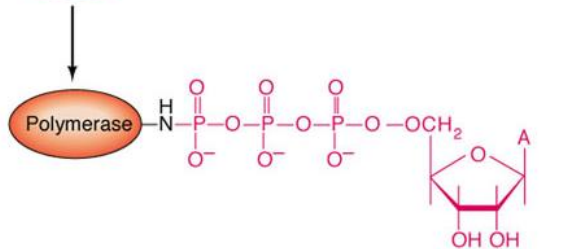


Reagent I

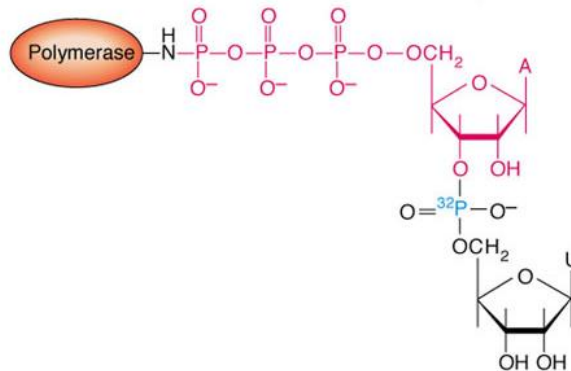
ATP analog

(b)

Reagent I +



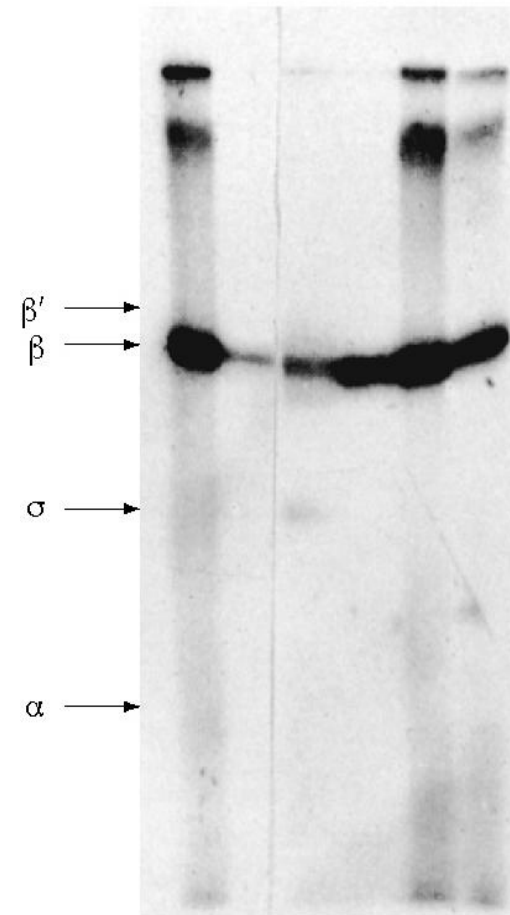
^{32}P -UTP



Affinity labeling experiment

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1 2 3 4 5 6



(Source: Grachev et al., Studies on the functional topography

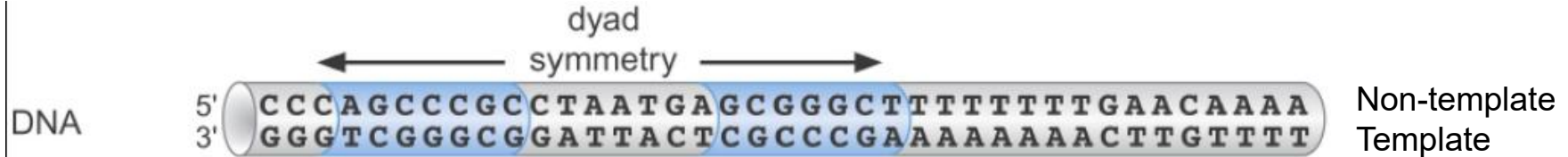
Summary: functions of α , β , β' subunits of RNA polymerase

- (1) All these three subunits are involved in the extension of RNA strand
- (2) α subunit can interact with the UP element of promoter to enhance the transcription
- (3) β are involved in the formation of phosphodiester bond
- (4) β' subunit Stimulates σ binding to core element of promoter

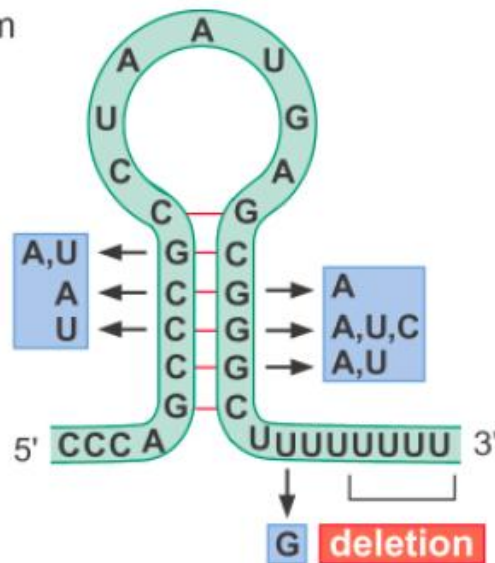
2.4 Termination of transcription

- Terminator: a specific DNA sequence that signals transcription to terminate
- ρ (rho) independent termination:
 1. An inverted repeat that allow a **hairpin** to form at the end of the transcription
 2. A string of Ts in the non-template strand that results in **a string of weak rU-dA** base pairs holding the transcription to the template strand.

Rho非依赖型终止子的序列



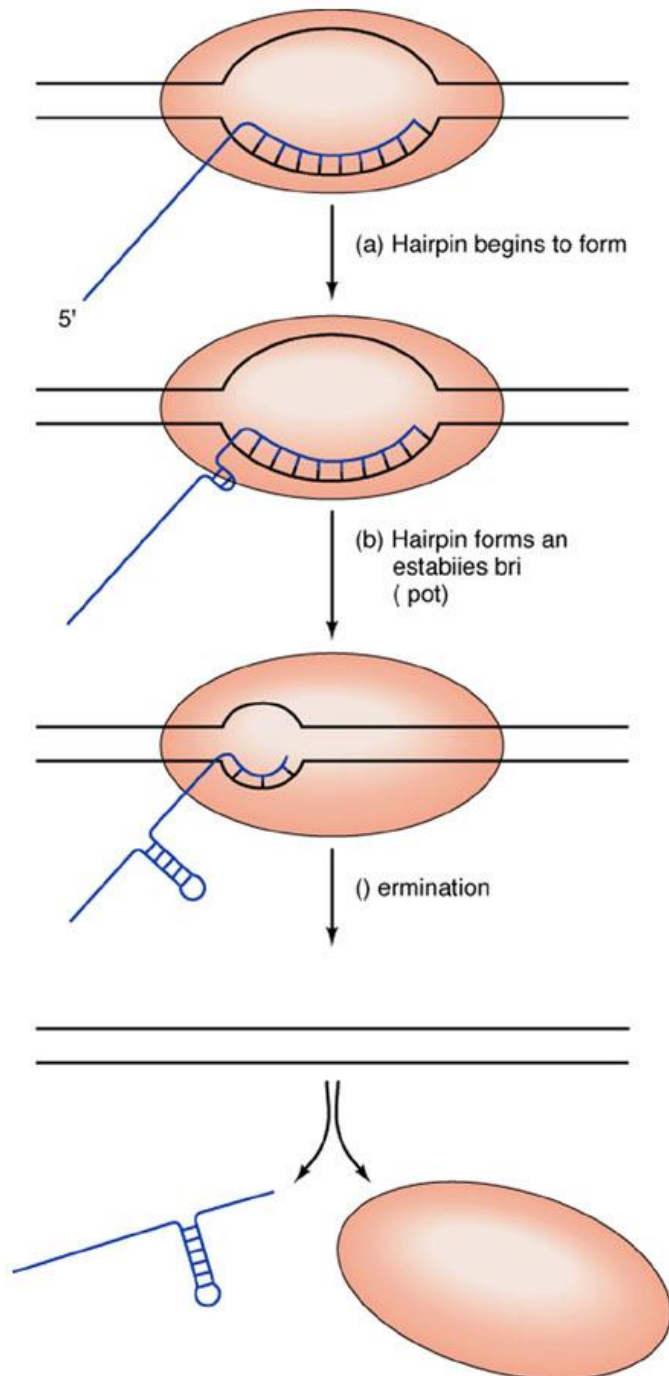
transcript folded to form termination hairpin



- 由两个序列元件组成：
- 一段短的反向重复序列（大约20个核苷酸）；
 - 其后是一段大约8个A：T碱基对的序列。



由它转录出mRNA可形成茎环结构，可阻止RNA pol的前进。



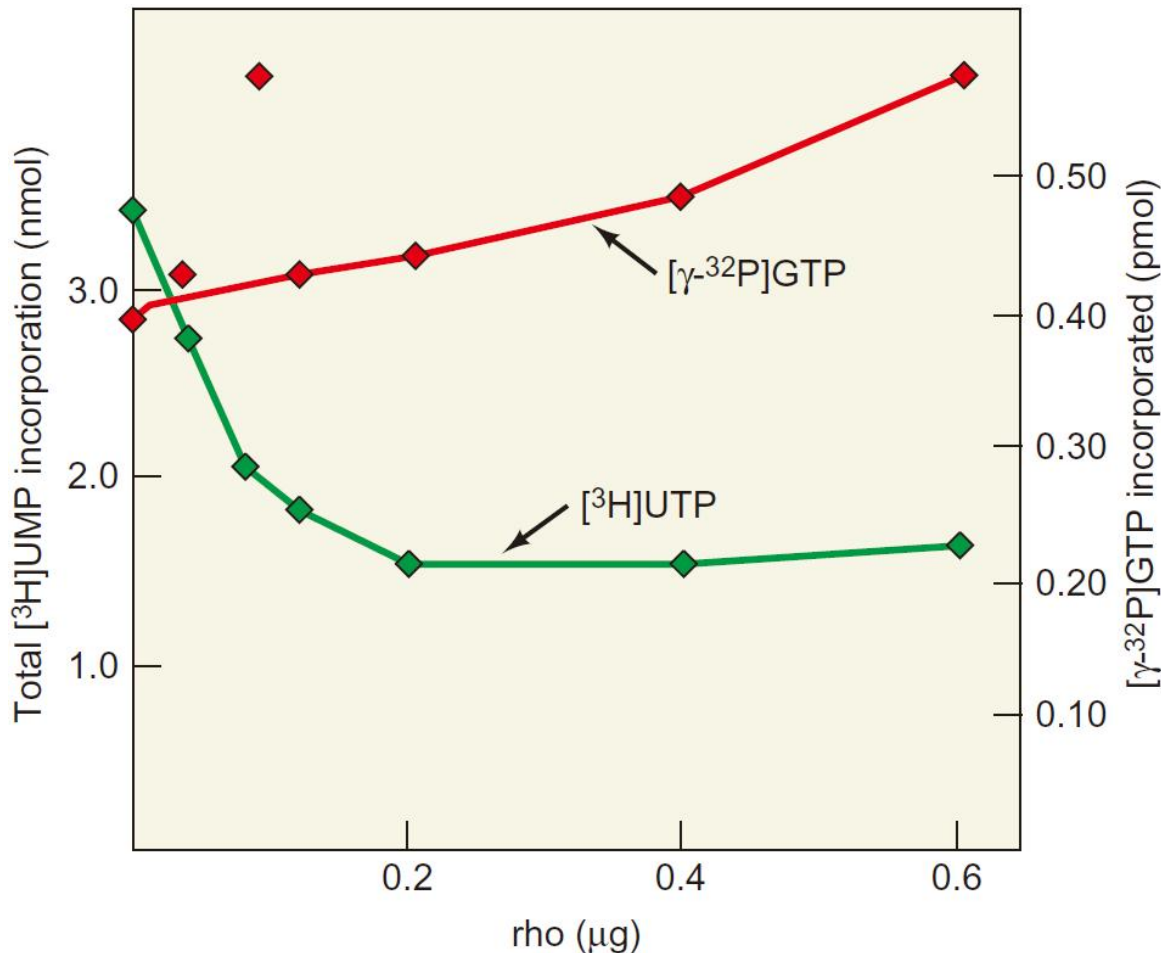
Model of ρ -independent termination

- (a) The polymerase has paused at a string of weak rU–dA base pairs, and a hairpin has started to form just upstream of these base pairs.
- (b) As the hairpin forms, it further destabilizes the RNA–DNA hybrid. This destabilization could take several forms: The formation of the hairpin could physically pull the RNA out of the polymerase; or it could simply cause the transcription bubble to collapse, expelling the RNA from the hybrid.
- (c) The RNA product and polymerase dissociate completely from the DNA template, terminating transcription.

ρ -dependent termination

- Need ρ factors
- Need the inverted repeat that allow a hairpin to form at the end of the transcription
- No need of a string of Ts in the non-template strand

ρ as a termination factor

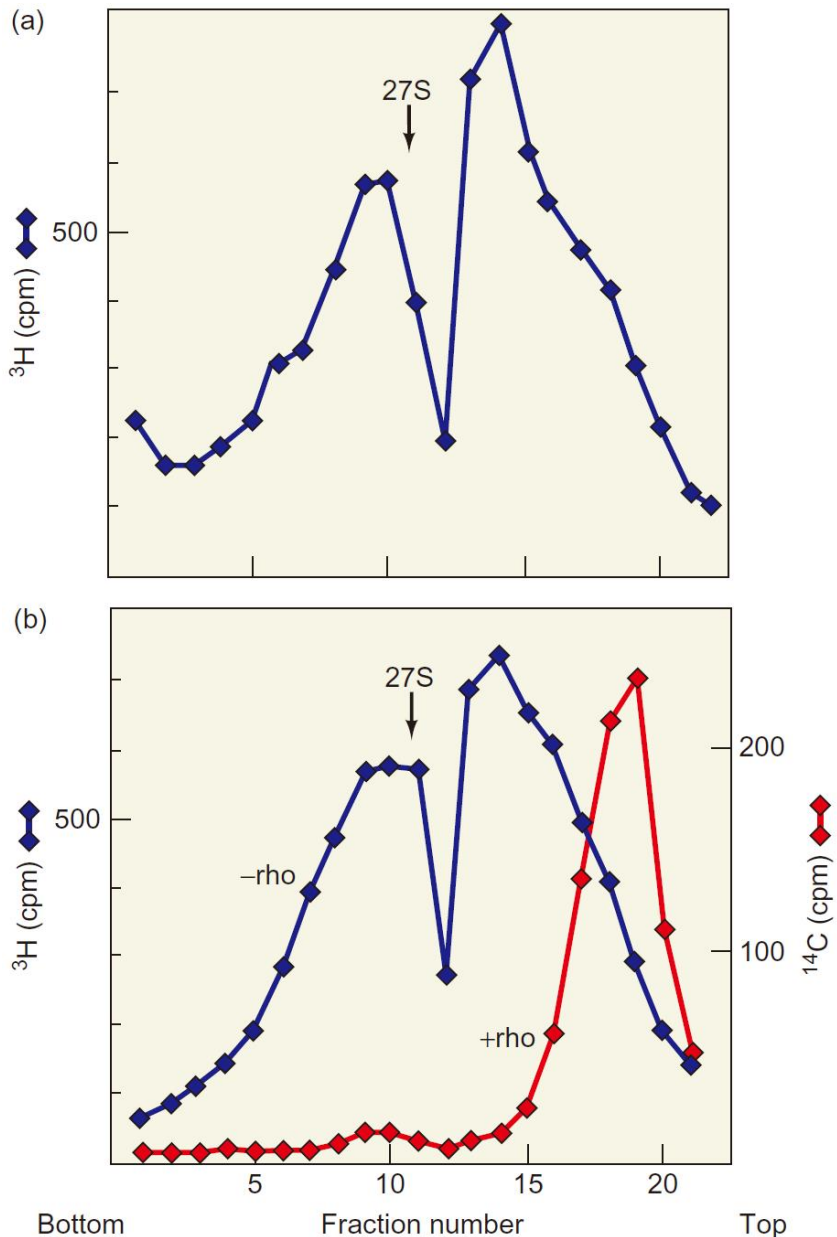


Rho decreases the net rate of RNA synthesis.

1. Roberts allowed *E. coli* RNA polymerase to transcribe λ phage DNA in the presence of increasing concentrations of rho.
2. [γ -³²P]GTP was used to measure initiation (red) and [³H]UTP to measure elongation (green).
3. Rho depressed the elongation rate, but not initiation.

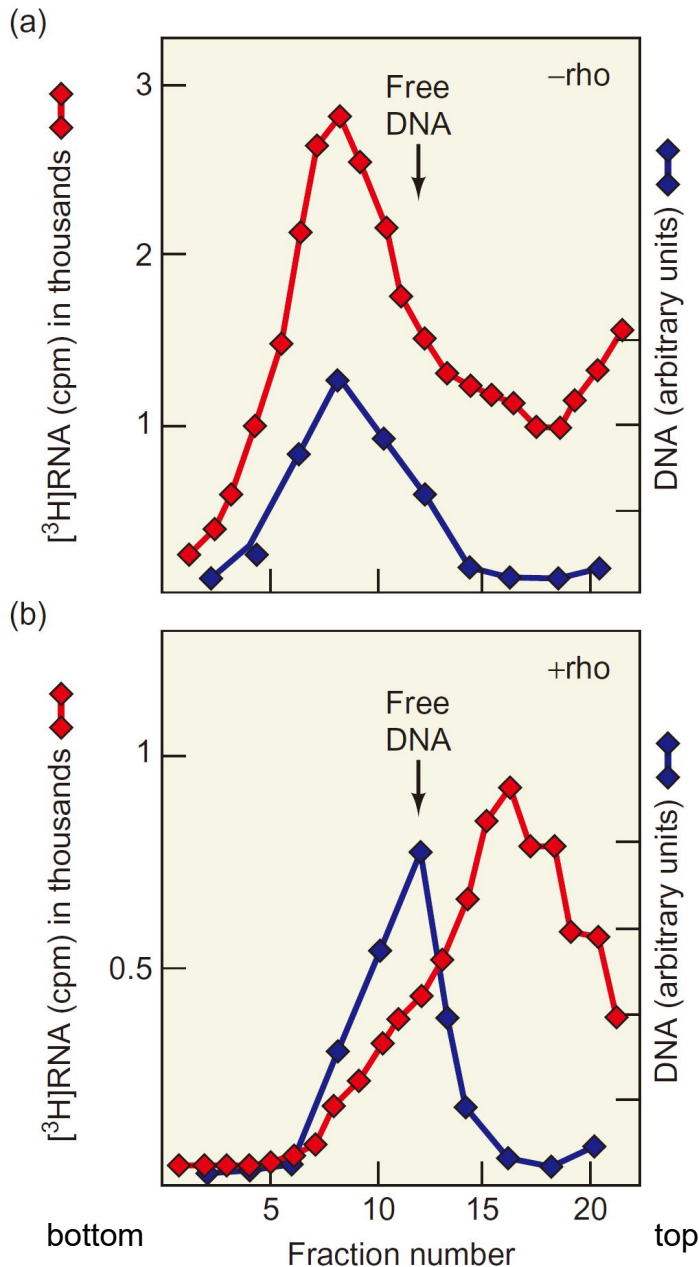
1969, Nature 224:1168-1174

ρ reduces the size of the RNA product



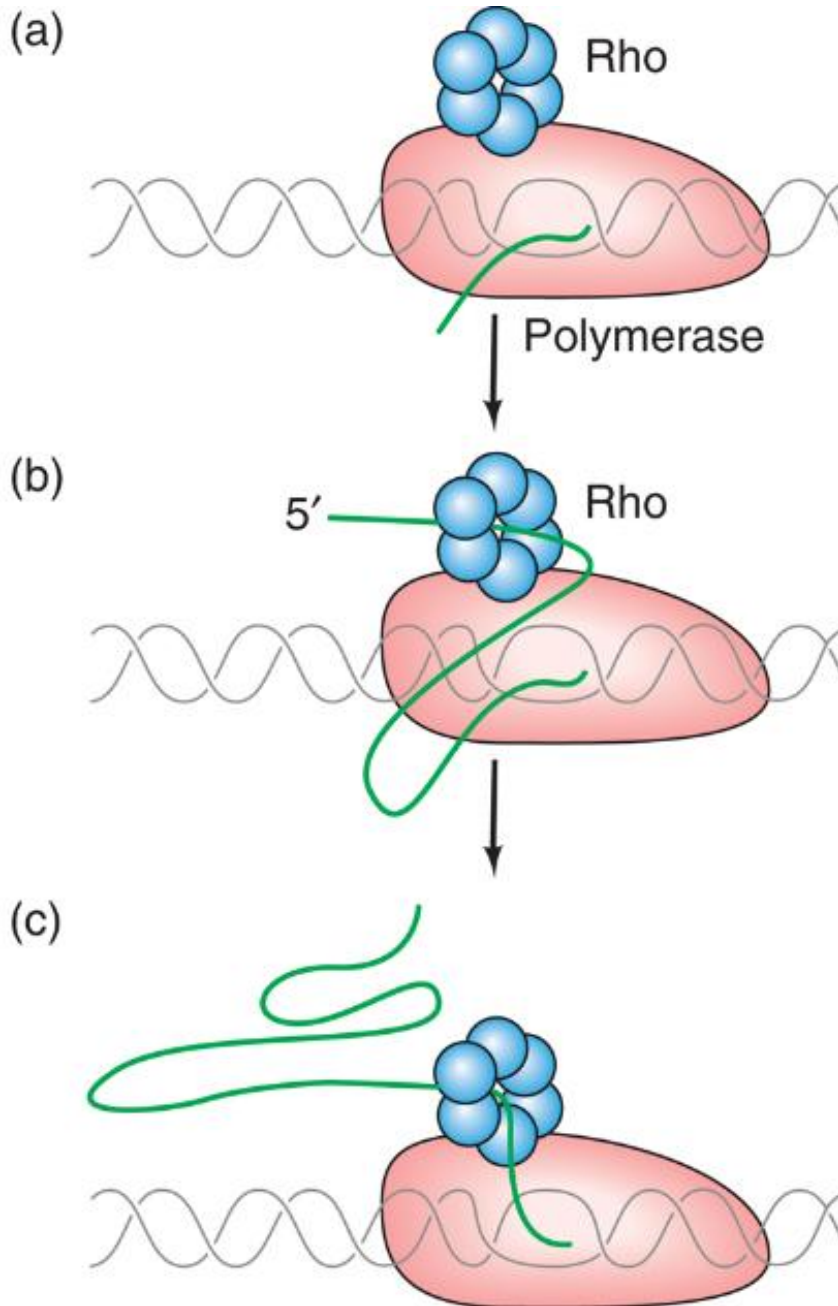
- *E. coli* RNA polymerase to transcribe λ DNA. [^{14}C]ATP was used to label the transcripts synthesized in the presence of rho. ^3H was used to label the transcripts synthesized in absence of rho.
- The ^{14}C -labeled transcripts (red) were relatively small, the ^3H labeled transcripts (blue) were originally.
- Thus, rho has no effect on the size of previously made transcripts, but it reduces the size of the transcripts made in its presence.

ρ Releases Transcripts from the DNA Template



- (a) absence or (b) presence of rho.
- ^3H -labeled product (red) to ultracentrifugation to see whether the product was associated with the DNA template (blue).
- (a) The RNA made in the absence of rho sedimented together with the template in a complex that was larger than free DNA.
- (b) The RNA made in the presence of rho sedimented independently of DNA at a position corresponding to relatively small molecules.
- Thus, transcription with rho releases transcripts from the DNA template.

The role of Rho- dependent termination



- (a) Rho has joined the elongation complex by binding directly to RNA polymerase. The end of the nascent transcript has just emerged from the polymerase.
- (b) The transcript has lengthened and has bound to rho via a rho loading site, forming an RNA loop. Rho can now feed the transcript through its central cavity.
- (c) The polymerase has paused at a terminator. By continuously feeding the transcript through itself, rho has tightened the RNA loop and irreversibly trapped the elongation complex. Rho has also begun to dissociate the RNA-DNA hybrid, which will lead to transcript release.

3. Regulation of transcription in prokaryotes

Control the expression of genes: some on, some off

Operon (操纵元)

Operon: a unit of gene expression in prokaryotes, which includes structural genes and control elements

- Each operon is subjected to stringent regulation:
on (increased transcription initiation) or off (decreased transcription initiation) depending on the needs of the microorganism.
- This allows the bacteria to be efficient: it will produce specific proteins only when they are required

Operator (操纵区): A DNA element found in prokaryotes that binds tightly to a specific repressor and thereby regulates the expression of adjoining genes

Typical operons

- Lac operon: utilization of the sugar lactose
- Ara operon: utilization of the sugar arabinose
- Trp operon: synthesis of the amino acid tryptophan

3.1 Lac operon

- **Three structure genes:**
 1. β -galactosidase (*lacZ*): break lactose into galactose and glucose
 2. Galactoside permease (*lacY*): to transport lactose into cells
 3. Galactoside transacetylase (*lacA*): unknown function, involving in metabolism of lactose

Structure of Lac operon



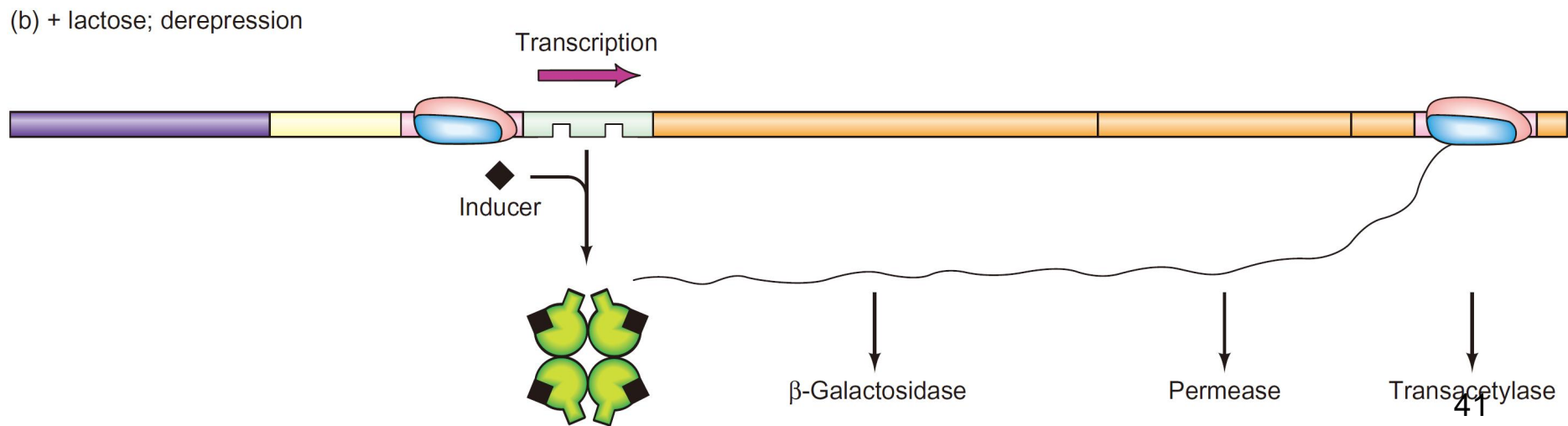
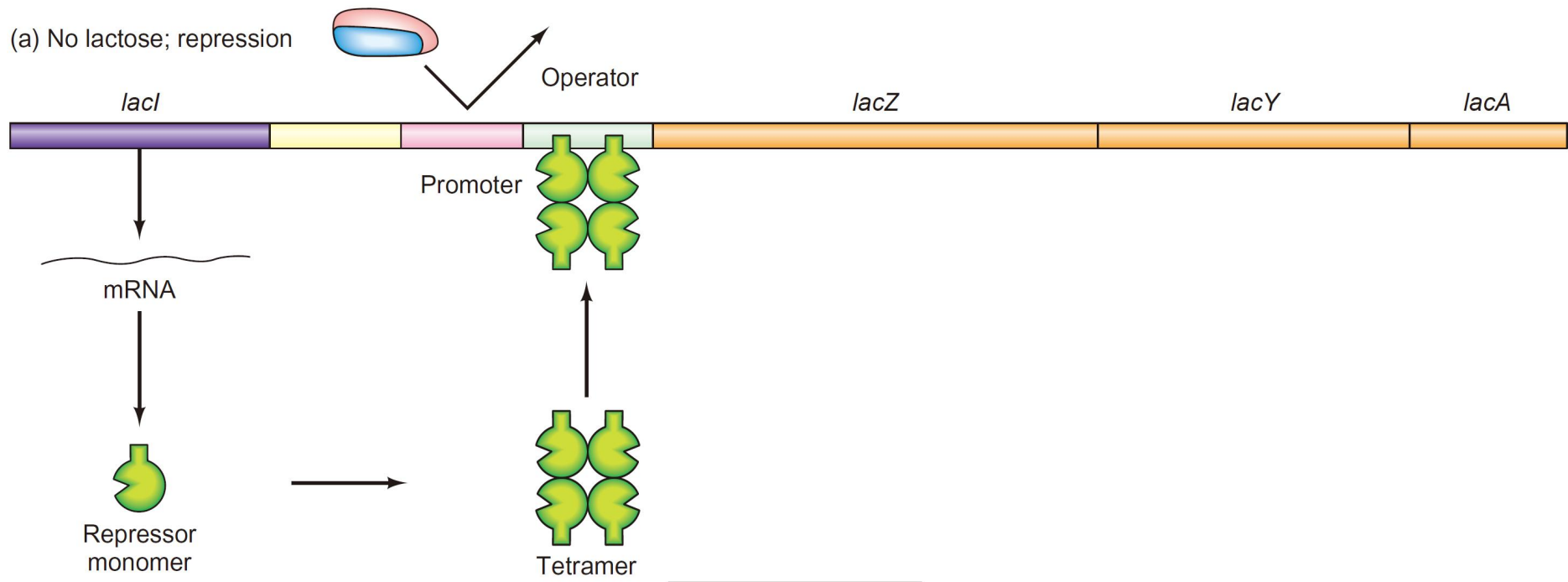
Lac I: Gene encoding the Lac repressor (LacR)

P: Lac promoter (binding site for RNA pol)

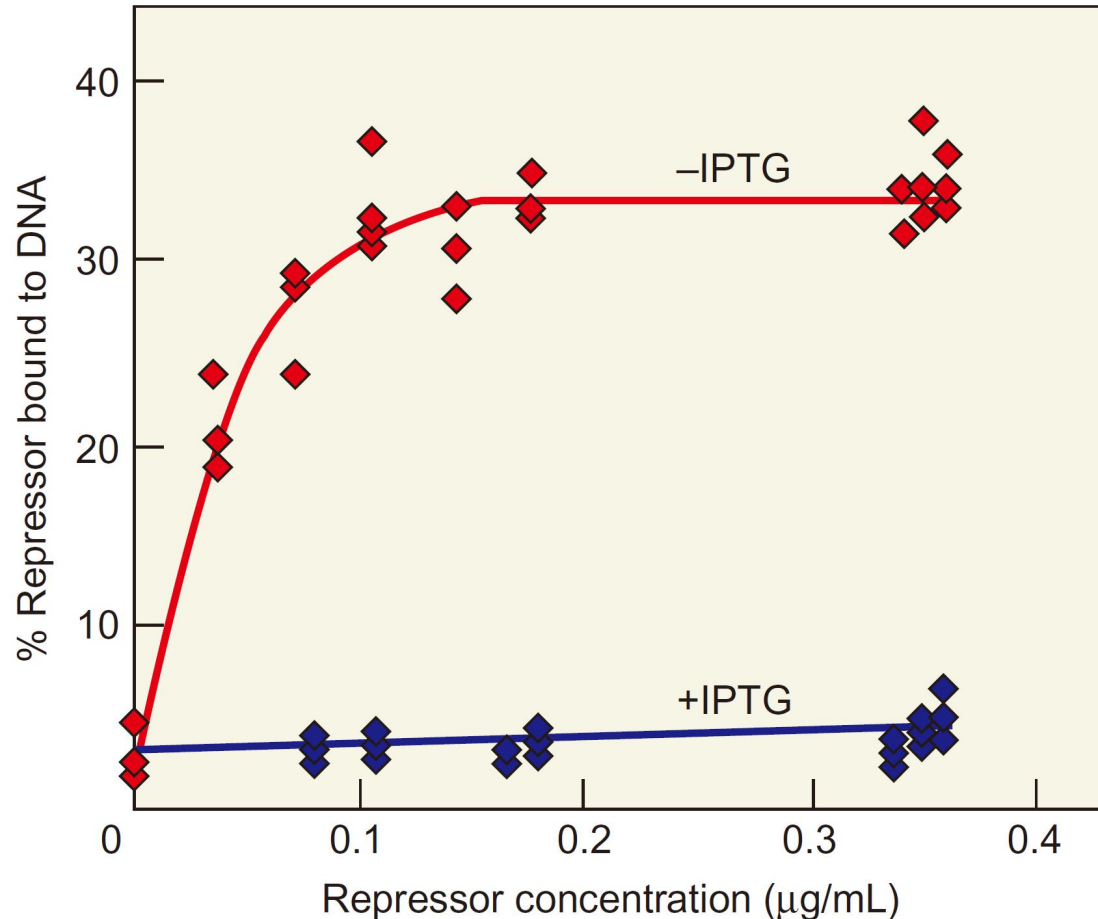
O: Lac operator (binding site for LacR)

- **Three structure genes:**

1. β - galactosidase (*lacZ*): break lactose into galactose and glucose
2. Galactoside permease (*lacY*): to transport lactose into cells
3. Galactoside transacetylase (*lacA*): involving in metabolism of lactose



Repressor-operator interactions



- **Labeled lacO-containing DNA with ^{32}p**
- **Binding b/w WT Lac operator & Lac repressor**

Summary of Negative control of the LAC operon

- **Two key elements:** repressor (阻遏物) and operator (操纵区)
- The operon is turned off as long as the repressor binds to the operator
- When the supply of glucose is exhausted and lactose is available, the few molecules of lac operon enzymes produce a few allolactose from lactose.
- The allolactose acts as an inducer by binding the repressor, repressor dissociate from the operator
- The operon now is turned on

Controls of the LAC operon

- Negative control:

The binding of repressor to the operator keeps the operon off.

- Positive control:

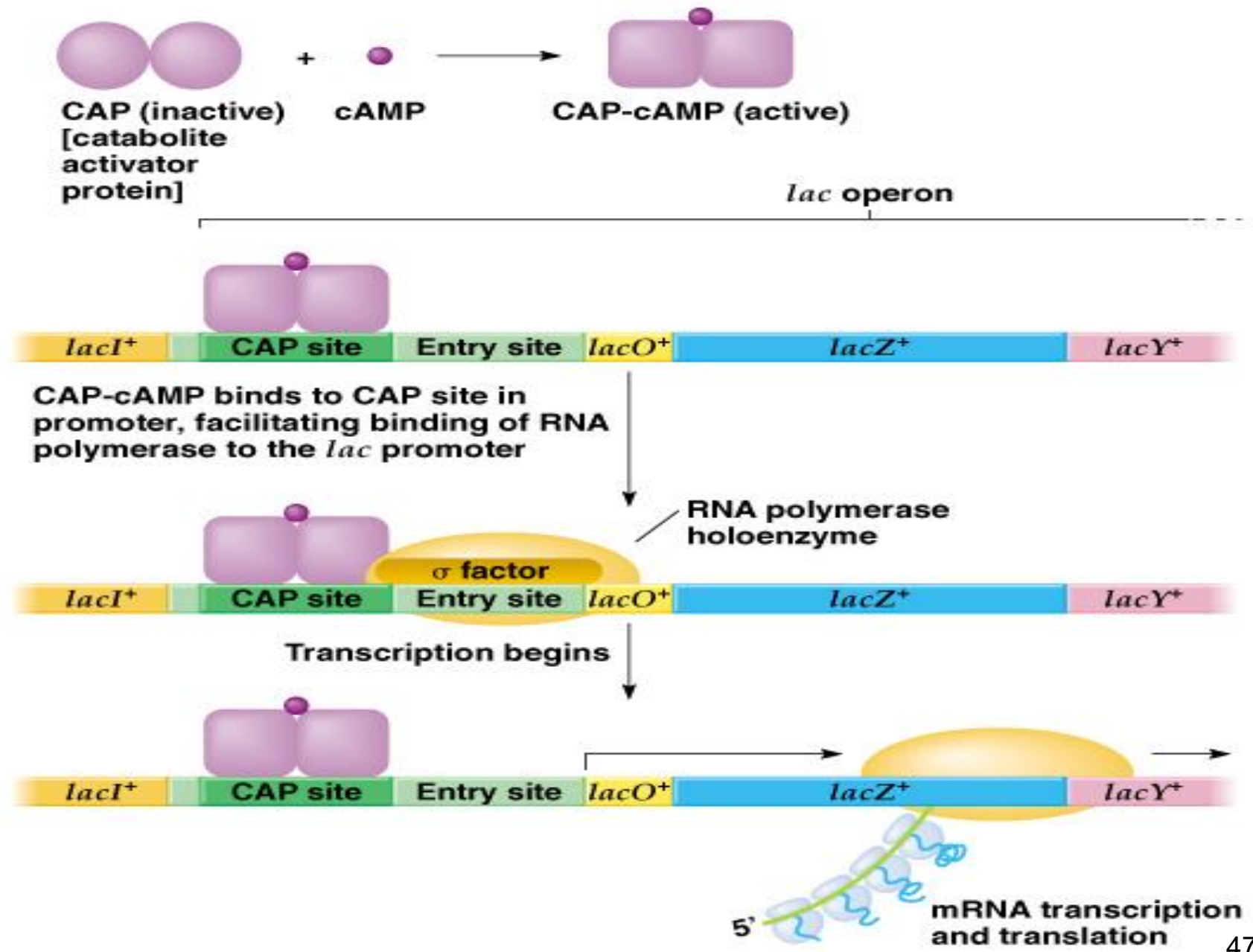
Catabolite activator protein (CAP) in conjunction with cyclic-AMP stimulates transcription.

Positive control of Lac operon

- Known:
 - selection in favor of glucose metabolism: with glucose, lac operon is off.
- Solution for the puzzle:
 - Who is response to glucose concentration? cAMP
 - Who is interact with cAMP & promoter?

Positive control of Lac operon

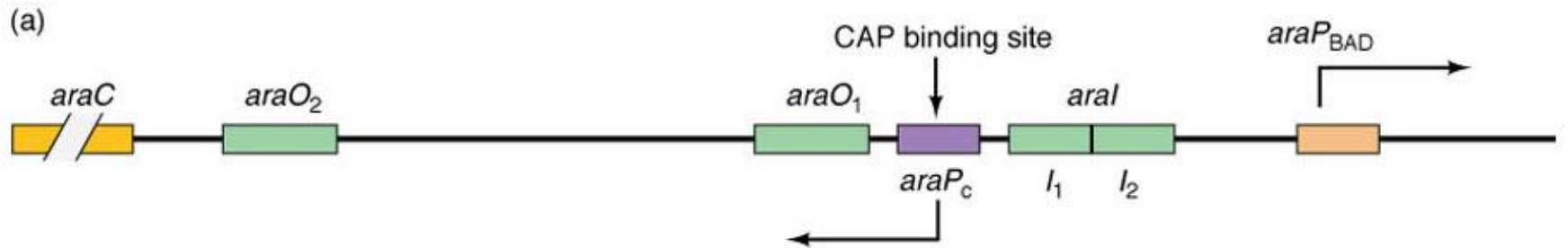
- Mediated by a factor called catabolite activator protein (CAP, 代谢分解物激活蛋白)
- CAP conjugated with cAMP (cyclic-AMP, 环腺苷酸) stimulate transcription
- cAMP concentration depends on glucose: more glucose, less cAMP
- The Lac operon is activated only when glucose concentration is low.



Summary of Lac operon

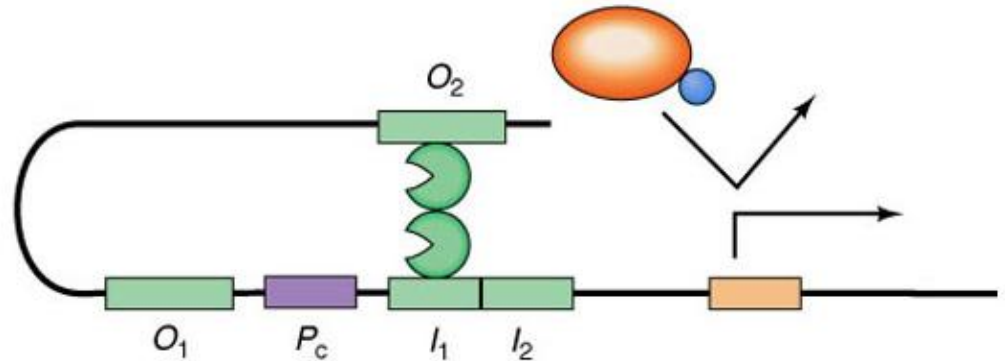
- Lac structure genes
only expressed
when:
 1. No Glucose
 2. Available of lactose

3.2 The ara operon

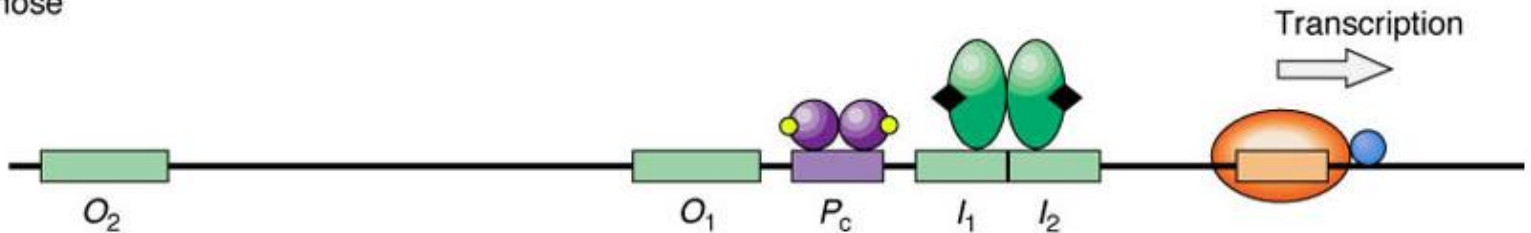


(b) – Arabinose

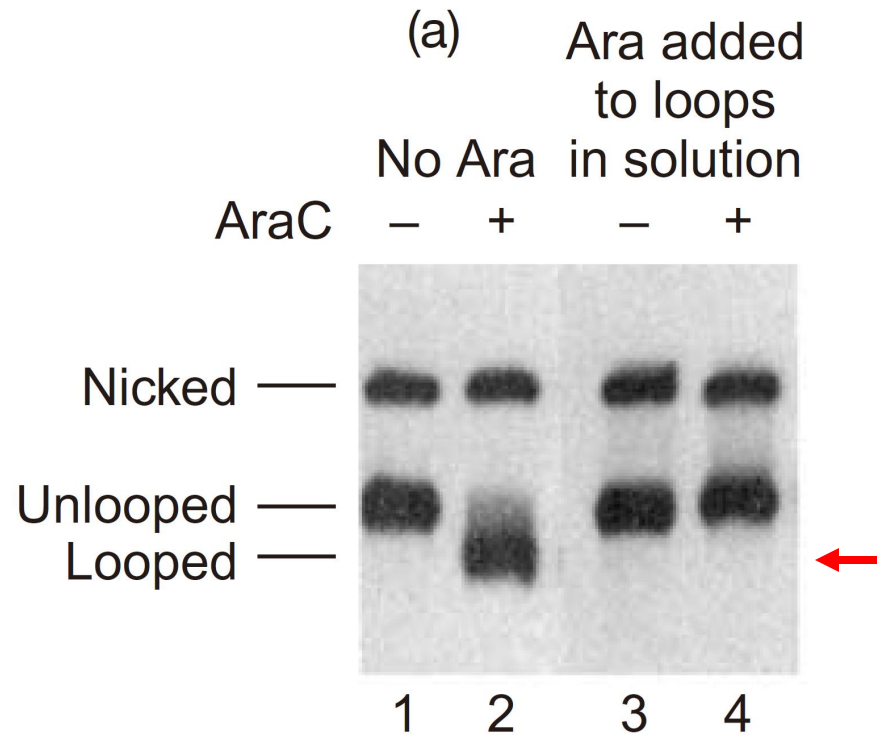
AraC protein as repressor, it binds to the operators in two sites and loop the 210bp DNA out



(c) + Arabinose



Arabinose acts as inducers



- **Supercoiled DNA have a higher electoretic mobility than the same DNAs unlooped**
- **Arabinose breaks the loop between araO2 and aral**

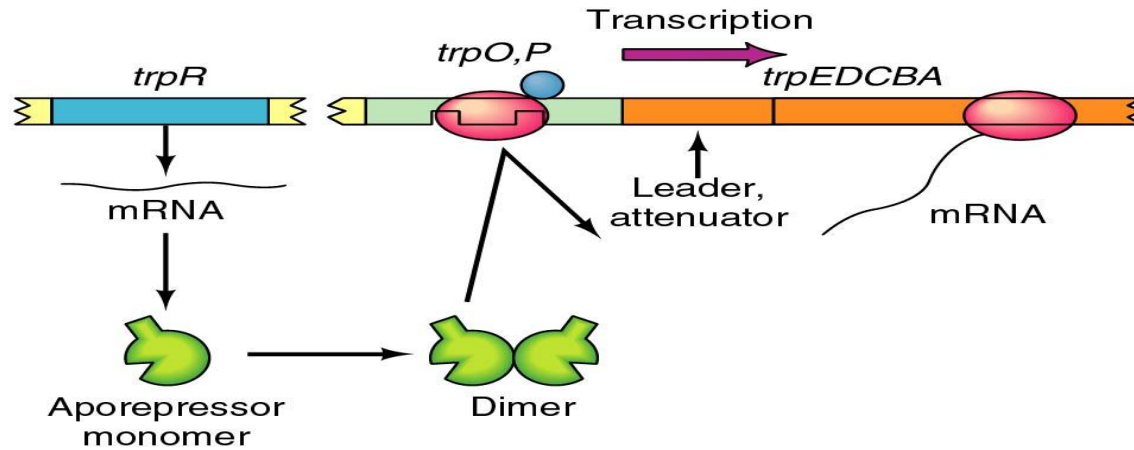
ara operon

- **Positive control by CAP-cAMP complex, which is similar to Lac operon**

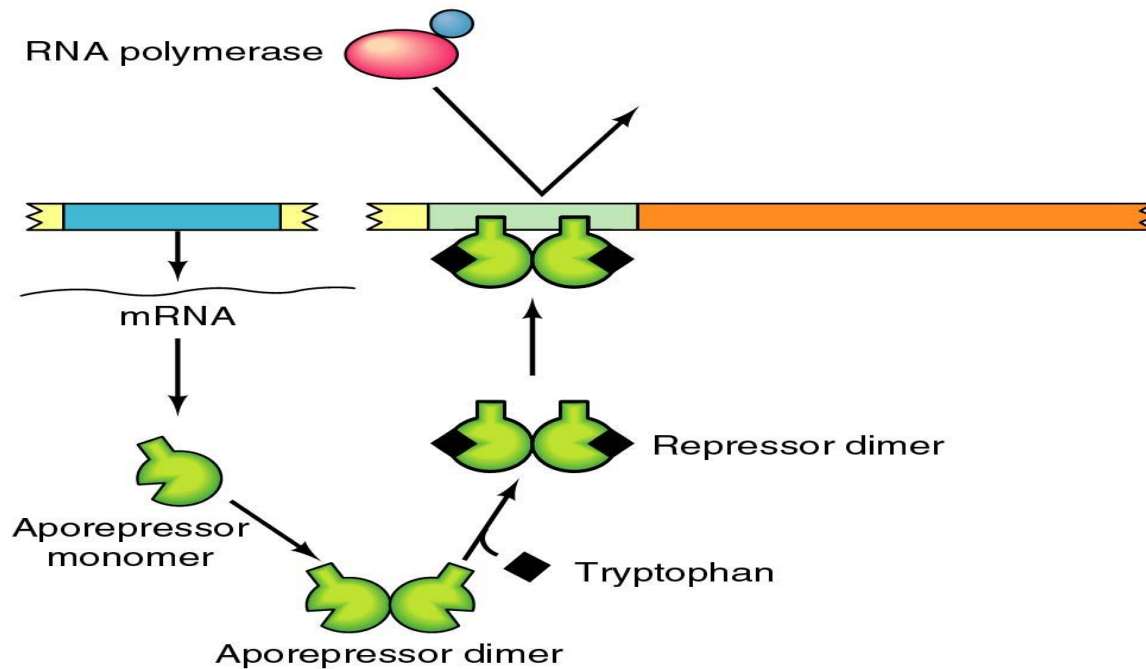
3.3 The *Trp* Operon (反馈抑制)

- Genes for enzymes making tryptophan
- Repression requires the binding of trp and apo-repressor (辅阻遏蛋白)
- Aporepressor is inactive, only when it binds to tryptophan, act as a repressor.
- Low trp, no repression
- High trp, repression

Low tryptophan: no repression



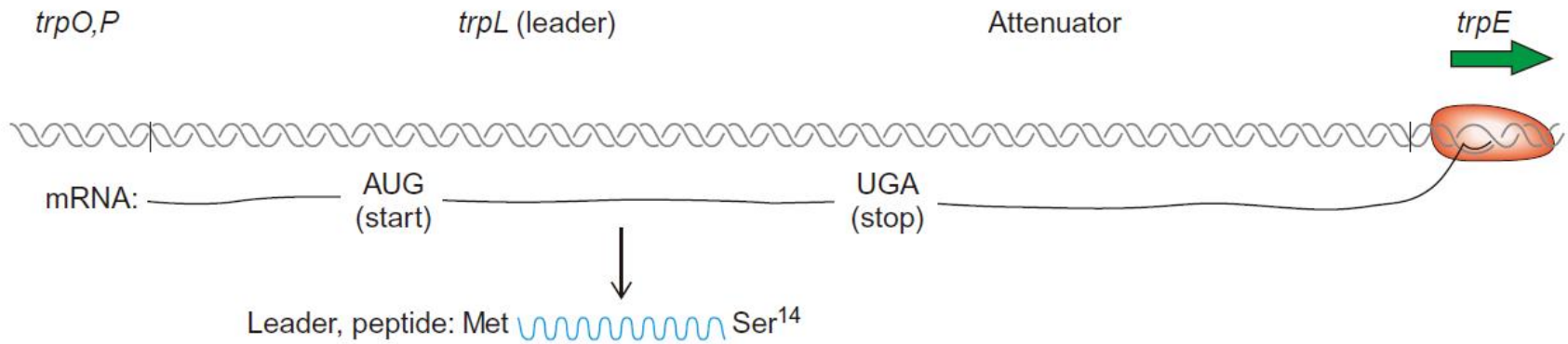
High tryptophan: repression



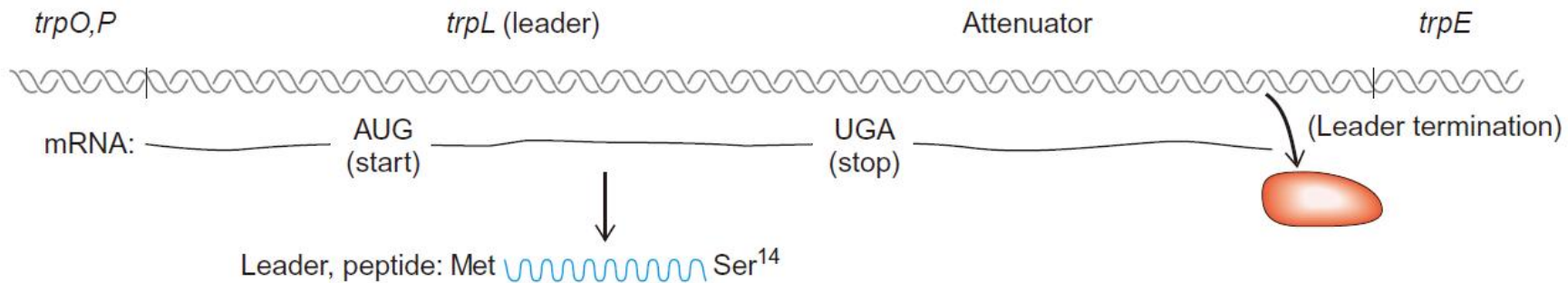
Control the trp operon by attenuation

- **Attenuator**: a region of DNA upstream of one or more structure genes, where premature transcription termination can occur
- Trp operon control 70 folds of transcription
- Attenuation controls additional 10 folds of transcription

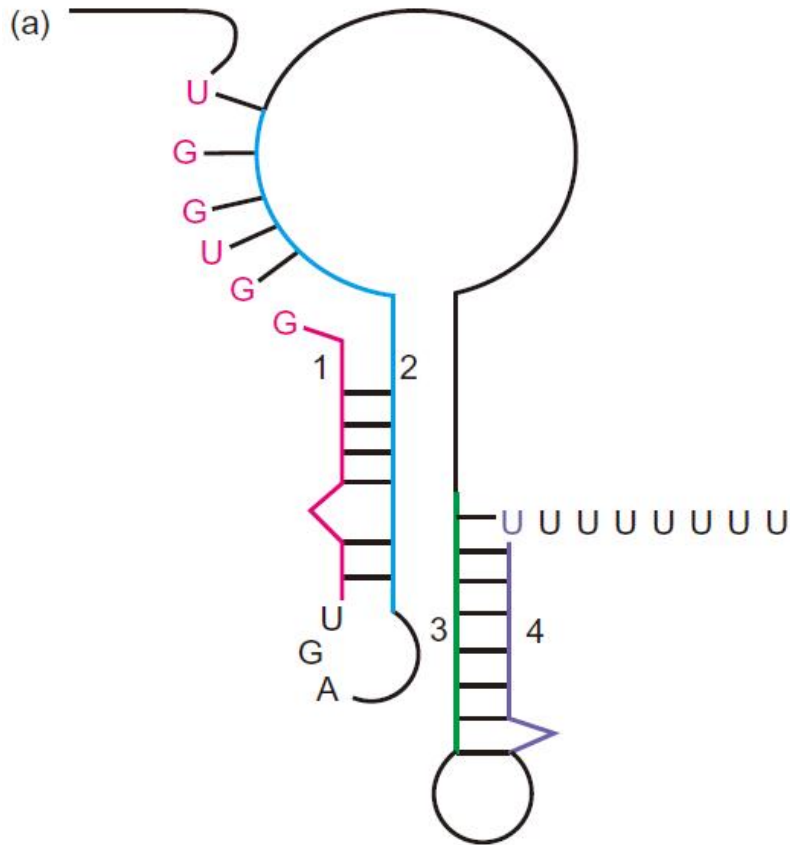
(a) Low tryptophan: transcription of *trp* structural genes



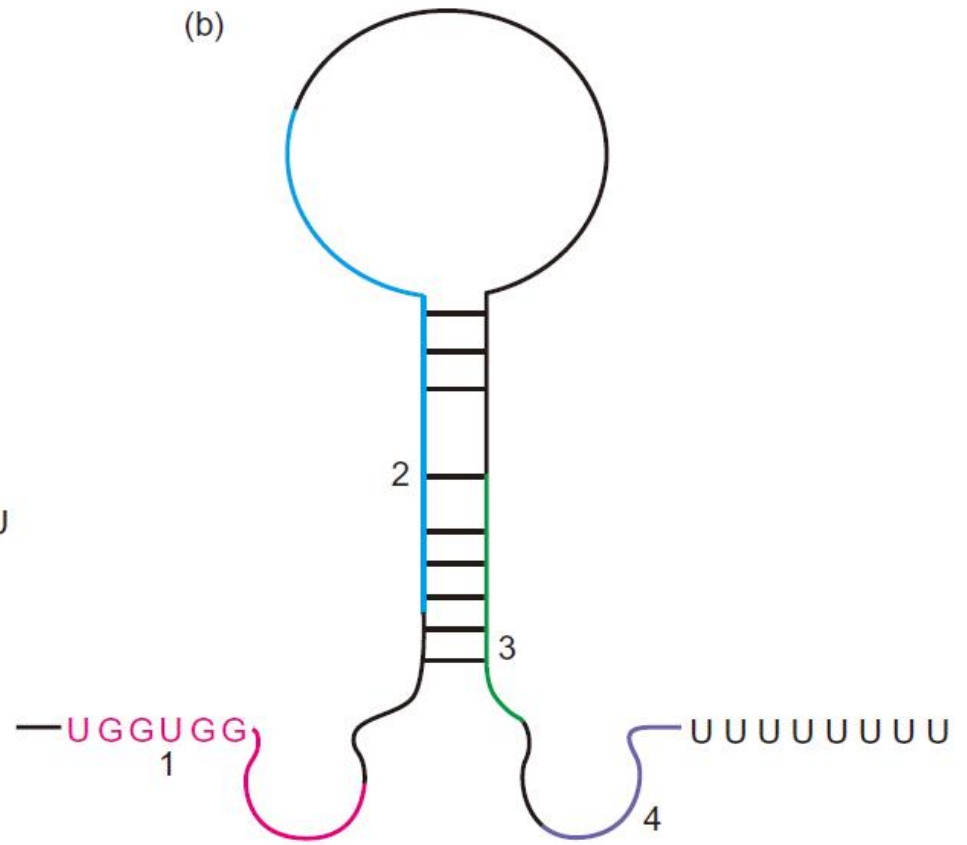
(b) High tryptophan: attenuation, premature termination



Two structures available to the leader–attenuator transcript

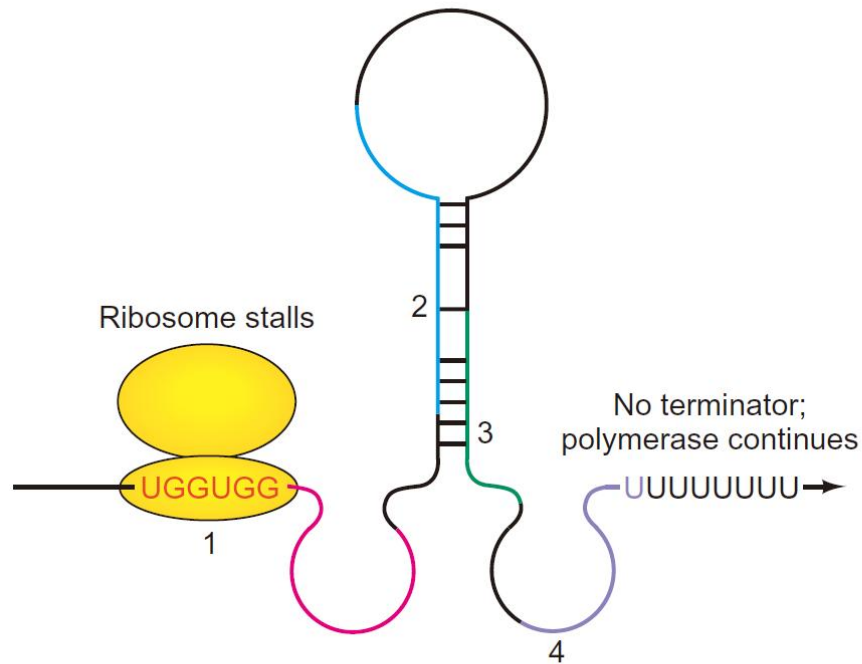


Looks like a terminator

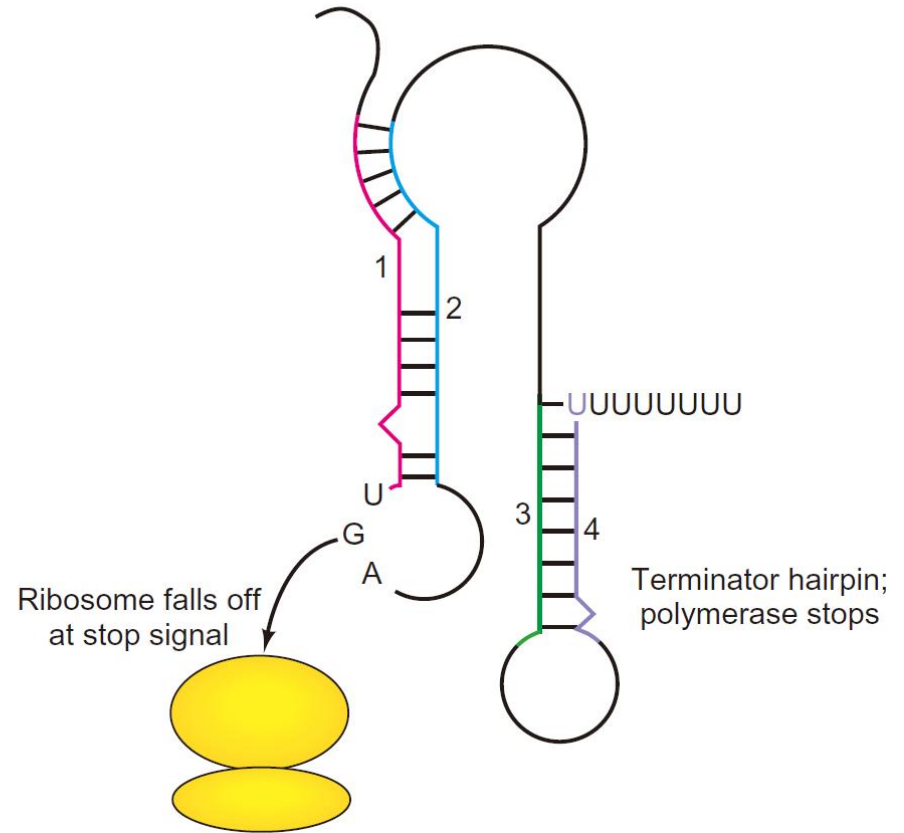


Does not look like a terminator

(a) Tryptophan starvation



(b) Tryptophan abundance



Met Lys Ala Ile Phe Val Leu Lys Gly Trp Trp Arg Thr Ser Stop
 pppA---AUGAAAGCAAUUUUCGUACUGAAAGGUUGGUGGCGCACUUCCUGA

Analytical questions:

? What would be the effect on trp operon attenuation if:

- There were no Trp codons in the leader peptide?

Always off

- There was No Trp tRNA in a cell?

Always on

- The attenuator sequence mutates to become AT rich?

Always on

Attenuation in other amino acid synthesis operons

- (A) Met - Lys - Arg - Ile - Ser - Thr - Thr - Ile - Thr - Thr - Thr - Ile - Thr - Ile - Thr - Thr - [Thr]
5' AUG AAA CGC AUU AGC ACC ACC AUU ACC ACC ACC AUC ACC AUU ACC ACA 3'
- (B) Met - Lys - His - Ile - Pro - Phe - Phe - Phe - Ala - Phe - Phe - Phe - Thr - Phe - Pro - Stop [Phe]
5' AUG AAA CAC AUA CCG UUU UUC UUC GCA UUC UUU UUU ACC UUC CCC UGA 3'
- (C) Met - Thr - Arg - Val - Gln - Phe - Lys - His - His - His - His - His - His - His - Pro - Asp - [His]
5' AUG ACA CGC GUU CAA UUU AAA CAC CAC CAU CAU CAC CAU CAU CCU GAC 3'

Leader peptides

Thanks!